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Healing Intervention for Improved the Efficiency and Stability of Tandem Devices on Industrial Textured Silicon

Wenchi Kong^{1,2†*}, Haowen Luo^{1†}, Bowen Yang^{3†}, Jiajia Suo^{3†}, Xuanpu Qu¹, Ruiyan Li⁴, Tianqi Wei¹, Jiajia Hong¹, Xinrui Han¹, Xuntian Zheng¹, Henan Feng¹, Lu Zhao¹, Zijing Chu¹, Mingliang Li⁵, Renxing Lin¹, Wenhua Zhang⁵, Hairen Tan^{1,2*}

¹National Laboratory of Solid State Microstructures, College of Engineering and Applied Sciences, Frontiers Science Center for Critical Earth Material Cycling, Jiangsu Physical Science Research Center, Nanjing University, Nanjing 210023, China

²Research and Development Center, Renshine Solar (Suzhou) Co., Changshu, Jiangsu 215513, China

³Department of Chemistry-Ångström Laboratory, Uppsala University, Uppsala SE-75120, Sweden

⁴Zhejiang Provincial Innovation Center of Laser Intelligent Equipment Technology, Wenzhou 325200, China

⁵National Center for International Research on Photoelectric and Energy Materials, School of Materials and Energy, Yunnan University, Kunming, 650091, China

[†]These authors contributed equally: Wenchi Kong, Haowen Luo, Bowen Yang, Jiajia Suo.

*Corresponding authors. E-mail: kongwenchi@nju.edu.cn; hairentan@nju.edu.cn

Abstract

Monolithic perovskite/silicon tandem solar cells are widely regarded as the next-generation, low-cost, high-efficiency photovoltaic technology. However, achieving high-quality wide-bandgap perovskite top cells on industrial textured silicon, particularly using hybrid two-step deposition methods that combine sequential co-evaporation and solution processing, remains a major challenge. In this work, we introduce a healing intervention strategy that significantly tailors perovskite film growth and enhances the film quality on textured silicon substrates. By applying an MASCN solution to heal the as-prepared films, which induces secondary crystal growth to enhance overall perovskite crystallization, we obtain a conformal perovskite layer characterized by columnar grains that extend through the full film thickness. The resulting 1.68 eV wide-bandgap perovskite solar cells achieve a champion efficiency of 21.1%, contributing to a certified stabilized tandem efficiency of 30.77% and an impressive V_{oc} of 1.915 V over an active area of 1.164 cm². Notably, an encapsulated device retains its initial performance after 3400 hours of continuous maximum power point tracking under one-sun illumination in ambient conditions, representing the excellent stability in perovskite/silicon tandem cells reported to date.

Introduction

Perovskite/silicon tandem solar cells will be rapidly developed, in an era where economic and resource constraints require the implementation of cost-effective multi-junction technologies¹⁻⁶. For tandem cells, the textured silicon with pyramids of a few micrometers in height not only has the advantage of being compatible with industry standards, but also has a greater photocurrent due to increased light trapping and reduced reflection losses, compared with nanotextured silicon⁷. However, the certified efficiency of the tandem cells by using hybrid two-step deposition method

based on the industry-standard textured silicon $>1\ \mu\text{m}$ is only 31.6% ($V_{oc}=1.925\ \text{V}$ $J_{sc}=20.30\ \text{mA}/\text{cm}^2$ $\text{FF}=80.9\%$), which is much lower than the efficiency of 34.58% ($V_{oc}=1.996\ \text{V}$ $J_{sc}=20.80\ \text{mA}/\text{cm}^2$ $\text{FF}=83.63\%$) based on the submicron-textured silicon $<1\ \mu\text{m}$ ^{1,3}. The significant difference of tandem cells mentioned above in PCE lies in film quality of the wide-bandgap perovskite top cells, which is critically dependent on the deposition method. More specifically, compared to the one-step solution method on textured silicon $<1\ \mu\text{m}$, the two-step, hybrid co-evaporation/solution approach used for perovskite deposition on textured silicon $>1\ \mu\text{m}$ often results in smaller grain sizes, poorer bulk quality, and inferior optoelectronic properties⁸⁻¹². It is well known that the non-radiative recombination losses within the bulk or at grain boundaries are the primary limitation for the state-of-the-art perovskite solar cells¹³⁻¹⁶. A major challenge arises because most crystallization-control strategies developed for flat glass substrates cannot be effectively translated to micron-textured silicon ($>1\ \mu\text{m}$). This limitation is largely due to the difficulty of introducing and regulating functional additives during the deposition of inorganic frameworks via evaporation processing and organic salt solutions via spin-coating or blade-coating methods on such complex surfaces. Therefore, developing strategies that tailor perovskite film growth and improve film quality throughout the full thickness on the micron-textured silicon is essential for addressing these challenges and achieving highly efficient tandem devices. Fortunately, it has been widely reported that SCN^- additive can promote crystallinity and grain growth on glass, demonstrating its critical role in regulating perovskite crystallization^{17,18}. However, how to effectively introduce SCN^- materials to fabricate large-sized grains and conformal perovskite films on the micron-textured silicon has never been reported.

Here, we employ a healing intervention strategy for perovskite top cells, in which an MASCN solution is applied to the as-prepared films to achieve conformal coverage on textured silicon. This approach enables the formation of high-quality perovskite films with minimal grain boundaries, low defect densities, and exceptional stability on various substrates through an Ostwald ripening process. Through in situ characterization of perovskite films, we elucidate the growth mechanism underlying both conformal coverage and enlarged grain formation achieved by the post-treatment strategy. Leveraging MASCN-based healing approach, we successfully fabricated tandem cells

with an active area of 1.164 cm² that achieve a certified PCE of 30.77% with a V_{oc} exceeding 1.915 V and maintain their initial performance at the maximum power point under one-sun illumination in air for over 3,400 hours.

Results

Formulation of healing intervention strategy

For commercial compatibility, the textured silicon with pyramids height greater than 1 μm is used for depositing perovskite film, and the corresponding surface and cross-sectional scanning electron microscopy (SEM) images of the silicon are shown in Supplementary Fig. 1a and b. The two hybrid co-evaporation/spin-coating strategies is used to deposit conformal perovskite films on textured silicon. In the co-evaporation step, lead iodide (PbI_2) and cesium bromide (CsBr) are deposited on silicon to form a conformal inorganic framework, as show in the SEM image in Supplementary Fig. 2a. In the second step, different organic salt and additives are employed, which is a key step in improving the quality of perovskite films. For example, spin-coating additive-free organic salt solution containing FAI/FABr in isopropyl alcohol (IPA) onto the inorganic framework, followed by annealing, we obtained perovskite films with small-sized grains, referred as-prepared film, as shown in the SEM image in Supplementary Fig. 2b.

In our previous work, we demonstrated that the addition of methylammonium thiocyanate (MASCN) into the FAI/FABr organic salt solution leads to the formation of super-large-grained perovskite films, yet same treatment results in exposed pyramid tips of the textured silicon substrate, as observed in Supplementary Fig. 3a and b¹⁹. However, the detailed processes and mechanisms responsible for this non-conformal grain coverage remained unclear. To address this issue, we elucidate the underlying growth mechanism through Laminar Flow and Level Set physics fields simulations in COMSOL model. Our analysis reveals that the FAI/FABr/MASCN solution rapidly penetrates the PbI_2/CsBr substrate, while MASCN simultaneously accelerates perovskite crystallization of perovskite films. This dual effect significantly increases the risk of perovskite grain depletion at the pyramid tips. Fortunately, the post-treatment engineering was always found to be effective in regulating the crystallization kinetics of the perovskite films on glass, controlling the surface morphology²⁰⁻²³. Therefore, we introduced a healing strategy and the result shows post-

treating the as-prepared perovskite film with MASCN solution prolongs the infiltration time and facilitates slow secondary growth, thereby reducing the exposure of pyramid tips. These dynamics are further detailed in Supplementary Note 1 and Supplementary Fig. 4-9. Since the perovskite films deposited from the FAI/FABr/MASCN solution fail to conformally cover the pyramid tips of the textured silicon—leading to current leakage and compromised device performance—we do not further discuss the optoelectronic properties of these films and the corresponding device performance in this study²⁴.

To suppress the exposure of the pyramid tips, we simultaneously incorporated both MASCN and methylammonium chloride (MACl) into the FAI/FABr solution, which is the optimal component with the highest PCE reported in our previous paper¹⁹. This strategy is based on the finding that MACl effectively retards perovskite crystallization via the formation of an intermediate phase. Therefore, the formulation containing both MASCN and MACl in FAI/FABr is denoted as the control in Fig. 1a, while healing intervention for the as-prepared perovskite film with MASCN solution is labeled as the target condition in Fig. 1b.

Impact of MASCN healing intervention as-prepared perovskite films

To capture the morphological evolution of the control and target perovskite films, we highlight their growth characteristics on the textured silicon substrate through a series of processing stages of the top-view SEM images. For the control film, deposition of the FAI/FABr/MASCN/MACl solution achieved conformal effect on textured silicon substrate. Then, a pre-annealing step was introduced, the wet perovskite film was annealed at 90 °C in a glovebox for 90 s to volatilize the IPA solvent, which facilitated continued grain growth in Fig. 2a and Supplementary Fig. 10a²⁵. However, the Cl⁻ in the mixed MASCN and MACl resulted in a reduction of grain size compared with single the additive MASCN shown in Supplementary Fig. 3a and b¹⁹. In contrast, deposition of MASCN solution onto as-prepared film resulted in a noticeable increase in grain size and a more conformal morphology in target, and the grain size further increases after pre-anneal in Fig. 2b and Supplementary Fig. 10b.

These observations reveal that the grain size during the MASCN deposition stage is substantially larger than that of the as-prepared film (Supplementary Fig. 2b), and continues to

increase after the pre-annealing stage, during which small-sized grains are almost completely eliminated. The grain growth of the as-prepared film is attributed to secondary crystallization induced by the MASCN solution, which drives via the Ostwald ripening mechanism (Supplementary Fig. 11), leading to improved film morphology. In order to gain an in-depth understanding of the crystallization and growth process of perovskite in the above based on PbI_2/CsBr and as-prepared film, we conducted X-ray diffraction (XRD) measurement. The (100) peak for the control film underwent a notably shift toward lower angle in Fig. 2c, indicating that chloride ions have entered the lattice or the substitution of small Br^- with large I^- . In contrast, the (100) peak of the target film showed negligible change in Fig. 2d, demonstrating that healing intervention as-prepared film by using MASCN solution primarily promotes grain growth and enhances crystallinity, SCN^- does not enter the lattice.

To further investigate Ostwald ripening process for large grain size and conformality of the perovskite film on textured silicon following healing intervention, we performed in situ photoluminescence (PL). A 405-nm laser was probed in a glove box to track PL of control and target films during nucleation and growth, the experiment was designed to collect the PL signal during the approximately 30 s of spin-coating, followed by annealing at 90 °C for 90 s. The result shows that the control perovskite undergoes a complex crystallization pathway in Fig. 2e. With prolonged spin-coating duration, gradual emergence of a perovskite phase exhibiting weak PL intensity is observed, concomitant with a progressive red-shift in the PL spectra, which reaches 692 nm after 30 s. For the target film, when MASCN is deposited on the as-prepared film, the PL wavelength does not change near 733 nm in Fig. 2f. However, both control and target leave only the perovskite phase with 728 nm after annealing. For the control film, annealing cause chloride ions to escape or the substitution of small Br^- with large I^- , thus resulting in a redshift of the PL peak position. While in the target film, the slight PL blue shift is attributed to the improved crystallization quality after pre-annealing in Fig. 2g²⁶. Among them, the half-peak width of the target is very narrow compared to control during the pre-annealing stage as observed in Supplementary Fig. 12a-d, indicating enhanced crystallization²⁷⁻²⁸.

The crystallization and optoelectronic properties of the final control and target films were also elaborated. The control exhibit stacked grains during 150 °C annealing in air (Supplementary Fig. 13a). The presence of numerous grain boundaries is a major contributor to non-radiative recombination losses and ion migration, thereby hindering device performance²⁹⁻³⁴. In contrast, the target film displays a more uniform structure, with columnar grains penetrating the film (Supplementary Fig. 13b). The enlarged grain size significantly reduces the density of grain boundaries, and the regrowth of grains effectively eliminates horizontal grain boundaries within the perovskite layer. To ensure that the MASCN healing intervention strategy does not disrupt the uniformity of the halogen distributions throughout the perovskite film, time-of-flight secondary-ion mass spectrometry (TOF-SIMS) measurement is performed. As observed in Fig. 3a and Supplementary Fig. 14a and b, the target film shows that the distribution of anions (Br^- , I^-) and cation (Cs^+) is as uniform as the control.

To investigate whether SCN^- ions remained in the both perovskite films, X-ray photoelectron spectroscopy (XPS) was performed. No S characteristic peaks were observed from the XPS spectra in Supplementary Fig. 15³⁵, verifying that SCN^- escaped from the film due to the thermal treatment. This measurement suggests that SCN^- is involved in the secondary grain growth but does not incorporate into the perovskite lattice. To further investigate the crystal orientation characteristics of the films on silicon, grazing incidence wide-angle X-ray scattering (GIWAXS) was performed. The control and target samples showed a Debye–Scherrer ring at $q_z = 1.0 \text{ \AA}^{-1}$ for the (100) plane of perovskite in Fig. 3b. Moreover, we integrated the diffraction intensity azimuthally over the above two Debye–Scherrer ring and observed the diffraction signals of the target film are more oriented and concentrated around azimuthal angles of approximately 45° and 135° in Supplementary Fig. 16. The orientation of grain growth is almost consistent with the pyramid slopes, which exhibit tilt angles near 53° and 127° (Supplementary Fig. 17 a and b), indicating promoted out of plane growth. These results reveal a clear contrast: the control film shows a randomly oriented distribution, whereas the target film—benefiting from SCN^- -induced orientation along the (100) plane—exhibits a highly ordered phase distribution and a simpler crystal structure, which further confirms its enhanced crystallinity. The control and target films

exhibited the low intensity of PbI₂ peak in Supplementary Fig. 18a, indicating slight decomposition of perovskite film during 150 °C annealing in air. Furthermore, the (100) peak intensity of the target film exceeds that of the control, and its full width at half maximum (FWHM) is reduced (0.103 vs. 0.167) in Supplementary Fig. 18b, signifying fewer crystal defects and improved crystallinity as evidenced by XRD. The preferential (100) orientation is expected to facilitate carrier transport across the perovskite films and their interfaces within photovoltaic devices³⁶.

Additionally, to elucidate the effects of target film on the optoelectronic properties of perovskite films, we conducted PL mapping, steady-state PL and time-resolved PL (TRPL) spectroscopy measurements. The PL mapping and spectra (Fig. 3c and d) reveal that the target film exhibits uniformity comparable to the control film at both the pyramid tip and valley of the silicon. Identical PL peaks at 730 nm confirm that the healing intervention strategy does not alter the bandgap of perovskite film. Meanwhile, the target film demonstrates significantly stronger PL intensity, consistent with its enhanced crystallinity^{37,38}, as evident from the XRD results. The target films/C₆₀ had higher PL intensity, confirming the reduced non-radiative recombination at the interface due to changes in the morphology of the perovskite film in Supplementary Fig. 19. Moreover, the carrier lifetime in target film extends from 240 ns for control to 400 ns (Fig. 3e), indicating reduced defect density and suppressed of bulk non-radiative recombination pathways^{39,40}. Furthermore, quasi-Fermi level splitting (QFLS) measurements for target films reveal a substantial reduction in bulk non-radiative losses compared to the control^{41,42}. Specifically, the non-radiative loss is reduced by 43 mV (Fig. 3f and Supplementary Table 1), highlighting the enhanced defect mitigation capability of SCN⁻ healing intervention. These significant QFLS variations underscore the critical role of bulk recombination in limiting device performance. Ultraviolet photoelectron spectrometry (UPS) measurements reveal that the Fermi level of the target film shifts upward (toward the conduction band) compared to that of the control film in Supplementary Fig. 20a-c, which is consistent with the observed increase in contact potential difference (CPD) mapping extracted from Kelvin Probe Force Microscopy (KPFM) in Supplementary Fig. 21a-d⁴³. This work function reduction of target film versus control is attributed to the suppression of recombination loss enabled by the low defect density on the perovskite

surface, which leads to an increase in surface carrier density and the formation of enhanced n-type nature that can contribute to the out-of-plane electron extraction and transports⁴⁴⁻⁴⁶. Beyond SCN^- , we found that the use of Cl^- solution and a combination of SCN^- and Cl^- also facilitates conformal growth and increased grain size in perovskite films on silicon based on as-prepared film. Top-view and cross-sectional SEM images (Fig. 2a and b, Supplementary Fig. 22a and b) reveal the grain size trend: target > $\text{SCN}^- + \text{Cl}^-$ > Cl^- > control. We attribute these results to stronger crystallization-promoting effect of SCN^- compared with Cl^- . Notably, the addition of Cl^- into organic salt solution alone does not enhance grain size, suggesting that anion-mediated healing intervention strategy plays a pivotal role in promoting grain growth and conformal coverage on silicon, thereby amplifying the functional impact of this material.

This healing intervention method extends to perovskite films deposited on glass substrates, achieving the same enhancement in grain size (Supplementary Fig. 23) and optoelectronic performance. GIWAXS analysis (Supplementary Fig. 24 and 25) confirms that SCN^- and $\text{SCN}^- + \text{Cl}^-$ induce a preferred (100) orientation in perovskite films on glass at X-ray incident angles of 0.05° and 1.0° , the azimuthal integration of (100) crystal planes showed that these diffraction signals are more oriented and concentrated at azimuthal angles around 90° and out-of-plane growth, mirroring the trend observed on silicon. Although Cl^- can also induce a preferred (100) crystal plane, these diffraction signals are randomly distributed. XRD measurements (Supplementary Fig. 26) further reveals that the SCN^- film on glass exhibits the narrowest FWHM of the (100) peak, indicating enhanced crystallinity. PL mapping results (Supplementary Fig. 27) show that target films exhibit the highest PL intensity and uniformity. Moreover, the target film on glass demonstrates the longest PL lifetime (1211 ns) and the highest quasi-Fermi level splitting (QFLS, 1313 mV), signifying minimal non-radiative recombination losses (Supplementary Fig. 28 and 29). These results indicate that the morphology, structure, and optoelectronic properties of the perovskite films healed with SCN^- are optimal, followed by $\text{SCN}^- + \text{Cl}^-$, Cl^- and control. Furthermore, the healing intervention strategy is broadly applicable for improving film quality, irrespective of the underlying substrate.

To evaluate the impact of the MASCN healing intervention strategy, single-junction solar

cells were fabricated. As shown in Fig. 3g, the target devices exhibit a clear improvement in average power conversion efficiency (PCE), increasing from $19.80 \pm 0.11\%$ to $20.91 \pm 0.13\%$. Notably, the open-circuit voltage (V_{oc}) increases from 1.186 ± 0.005 V to 1.232 ± 0.004 V, and the fill factor (FF) improves from $80.31 \pm 0.33\%$ to $81.85 \pm 0.29\%$. A champion PCE of 21.1%, with a V_{oc} of 1.237 V under reverse scan in Fig. 3h, was achieved—underscoring the effectiveness of the healing intervention strategy in enhancing perovskite solar cell performance.

Photovoltaic performance and stability of tandem cells

Lastly, we applied this strategy to fabricate monolithic perovskite/silicon tandem solar cells with the structure Ag/ITO/Si/ITO/ NiO_x /SAM/WBG perovskite/ C_{60} /SnO₂/IZO/Ag/MgF₂ (Fig. 4a). Among them, the PCE of the silicon solar cell is 21.94% in Supplementary Fig. 30. The SEM cross-sectional image of the target tandem device is shown in Fig. 4b. The WBG perovskite films exhibited uniform morphology and conformal growth on 2-3 μm silicon pyramids. These films featured vertically aligned columnar grains extending through the full thickness without grain boundaries, while the lateral grain size was significantly larger than that of the control tandem cells (Supplementary Fig. 31). Notably, the pyramid tips—typically the most challenging regions to achieve complete coverage—was seamlessly occupied by a single grain without grain boundaries in target device. This substantial reduction in grain boundaries and the uniform coverage of grains is a critical advancement for enhancing both the efficiency and stability of tandem solar cells, which can be observed from the photovoltaic parameters statistics results in Fig. 4c. Compared with the control tandem cells, the average V_{oc} improved from 1.86 V to 1.90 V for target. It is worth emphasizing that the FF has also been significantly improved alongside the V_{oc} improvement, which is associated with more efficient charge extraction at C_{60} interface, the suppressed non-radiative recombination and the reduced interfacial contact resistance. The average PCE achieved from 13 batches is 30.7%, demonstrating the reproducibility of high-efficiency tandem cells in Supplementary Fig. 32 a and b. The forward and reverse J - V scans of the best-performing target tandems are presented (the reverse scan with a PCE of 30.71, V_{oc} of 1.900 V, J_{sc} of 20.22 mA/cm^2 and FF of 79.93%) in Supplementary Fig. 33. The SIMIT-certified forward and reverse J - V scans of the unencapsulated target tandem device (Fig. 4d) display negligible

hysteresis and confirm our in-house measurements, achieving a reverse-scan PCE of 30.77%, with a J_{sc} of 20.13 mA/cm², a V_{oc} of 1.915 V, and an FF of 79.85%. The certification report is shown in Supplementary Fig. 34. The external quantum efficiency (EQE) measurement of the target device (Fig. 4e) indicates that both subcells produced an integrated J_{sc} >20 mA/cm², which is in agreement with the J - V measurement. A certified stabilized efficiency of 30.51% was realized under maximum power point (MPP) tracking for over 300 s (Fig. 4f). To assess long-term operational stability, encapsulated target tandem cells were tested under MPP using AM 1.5G one-sun illumination at room temperature in air. An encapsulated target tandem device exhibits excellent stability, maintaining their initial PCE of 28.4% without noticeable degradation over 3,400 hours (Fig. 4g). This impressive durability is primarily attributed to the reduced grain boundary density, which effectively suppresses halide ion migration.

Discussion

Through Laminar Flow and Level Set physics fields simulations in COMSOL model, we elucidate the growth mechanism underlying both conformal coverage and enlarged grain formation achieved by the healing intervention strategy. Perovskite layer based on MASCN-healing intervention is conformal and penetrated by one columnar grain on micron-textured silicon. This mitigated non-radiative recombination losses that occur in bulk perovskites on textured silicon and improve (100) orientation. We obtained efficient tandem solar cells with a V_{oc} exceeding 1.915 V and a PCE of 30.77%. The target tandem cells retain the performance no degradation after 3400 h at the maximum power point under one-sun illumination in air due to the greatly reduced grain boundaries. This study presents an effective technique for enhancing the quality of perovskite films fabricated via a hybrid two-step deposition method, thereby addressing the photovoltage-loss-induced efficiency limitation in industrial silicon-based tandem solar cells.

Methods

Materials

All materials were used as received without further purification. The organic salts (FAI, FABr, MASCN and F-PEACl, CF₃-PACl with the purity of >99%) were purchased from GreatCell Solar Materials. MACl and ethylenediamine diiodide (EDAI₂) was purchased from Xi'an Polymer Light Technology. PbI₂ (99.99%, trace metals basis) and CsBr (99.9%), 2PACz (>98.0%) and MeO-2PACz (>98.0%) were purchased from Tokyo Chemical Industry (TCI). IPA (99.5% anhydrous) were purchased from Sigma-Aldrich. The C₆₀ was purchased from Nano-C. MgF₂ (99.999%) was purchased from ZhongNuo Advanced Material Technology Co., Ltd. Tetrakis(dimethylamino) tin(iv) (99.9999%) was purchased from Nanjing Ai Mou Yuan Scientific Equipment. ITO glass has a square resistance of 20 ohm, purchased from SuZhou ShangYang Solar Technology.

Organic salt solution

In the fabrication of single-junction solar cells, for control, FAI (2.8 mmol), FABr (2.8 mmol), MACl (0.7 mmol) and MASCN (0.7 mmol) were dissolved in 15 ml IPA. For target, FAI (2.8 mmol) and FABr (2.8 mmol) were dissolved in 15 ml IPA. MASCN (0.7 mmol) dissolved in 15 ml of IPA was used to heal the as prepared films via spin coating at 4000 r.p.m. for 30 s within a nitrogen filled glovebox. The coated films were thereafter transferred to a hotplate and subjected to a pre-annealing step at 90 °C for 90 s under N₂ atmosphere, followed by a final annealing treatment at 150 °C in ambient air (35% relative humidity) for 20 min. In the fabrication of tandem solar cells, the concentration of the organic salt for fabricating single-junction solar cells was increased by 1.5 times.

Fabrication of single-junction perovskite solar cells

Patterned ITO glass substrates were sequentially cleaned using ultrasonication in acetone and isopropanol for 30 min, respectively. The cleaned ITO substrates were treated with UV ozone for 15 min, followed by sputtering a NiO_x hole transport layer on the substrates using a 3.5-inch NiO_x target (Plasmaterials, 99.9%) at 150 °C in an Ar atmosphere. The NiO_x layer was sputtered at a rate of 0.2 Å/s using a 90 W of RF power. Subsequently, the solutions of 2PACz and MeO-2PACz with the same concentration (1 mmol/ml in IPA) were mixed with volume ratios (75:25) and then were spin-coated on the NiO_x film at 4000 r.p.m. for 30 s, followed by annealing at 100 °C for

5 min in air to form molecule bridged interfaces, followed by washing with IPA for 2 times. The perovskite was produced using a two-step process. First, PbI_2 and CsBr were co-evaporated by thermal evaporation. The evaporation rate of PbI_2 and CsBr was 4 and $0.8 \text{ \AA s}^{-1}$, respectively. The thickness of the inorganic frame is 405 nm on a glass substrate and 540 nm on a textured silicon substrate.

Following, the organic salt solution with or without additives (MASCN or MACl) was spin-coated onto the inorganic films at 4000 r.p.m for 30 s in glovebox. Next, the films were transferred on a hotplate and pre-annealed at 90°C for 90 s in N_2 , followed by annealing at 150°C in ambient air (35% RH) for 20 min. For the device with passivation, the substrates were spin-coated 1mg/ml F-PEACl and $\text{CF}_3\text{-PACl}$ mixed solution with 4000 r.p.m. for 30 s and annealed at 100°C for 10 min. Sequentially, 20 nm of C_{60} film was deposited by thermal evaporation at a rate of $0.2 \text{ \AA s}^{-1}$. The substrates were then transferred to the ALD system (Veeco Savannah S200) to deposit 15 nm SnO_x at low temperature (typically 100°C) using precursors of tetrakis (dimethylamino) tin(IV) and deionized water. After that, 150 nm of Cu as an electrode was deposited by thermal evaporation at the rates of $1.0 \text{ \AA s}^{-1}$.

Silicon bottom solar cell fabrication

Silicon heterojunction (SHJ) bottom cells were fabricated on n-type double-side textured silicon wafers (CZ, 4 inch, $1\text{--}5 \Omega$ resistivity, 260 μm thickness). The intrinsic, n-type and p-type hydrogenated amorphous silicon layers were deposited on both sides of the wafer using plasma-enhanced chemical vapor deposition (PECVD). The thickness of back and front ITO was 100 nm and 20 nm in the silicon substrate for fabricating tandem cells. Micron-thick Ag is thermally evaporated on the back.

Perovskite/silicon monolithic tandem fabrication

Before the fabrication of perovskite top cells, the 4-inch silicon wafers were laser cut to $2.5 \times 2.5 \text{ cm}^2$ substrates. Then the silicon substrates were annealed at 200°C for 10 min to recover the sputtering damage.

The perovskite top cells were fabricated using the identical process of the single-junction perovskite solar cells while the thickness of inorganic frameworks were increased to 540 nm. After

depositing SnOx, 50 nm of IZO was sputtered and then Ag (400 nm) were deposited by thermal evaporation at the rates of $1.0 \text{ \AA s}^{-1}$. Finally, 100 nm of MgF₂ as an antireflection layer was deposited by electron beam evaporation.

Characterization of solar cells

For tandem solar cells, the current density-voltage (J - V) characteristics were measured using a Keithley 2400 source meter under the illumination of the solar simulator (EnliTech, Class AAA) at the light intensity of 100 mW cm^{-2} as checked with NREL calibrated reference solar cells (KG-5 and KG-0 reference cells were used for the measurements of wide-bandgap and narrow-bandgap solar cells, respectively). Unless otherwise stated, the J - V curves were all measured in air with a scanning rate of 100 mV s^{-1} (voltage steps of 20 mV and a delay time of 100 ms). The active area was determined by the aperture shade masks (0.049 cm^2) placed on the glass side of the solar cells. EQE measurements were performed in ambient air using a QE system (EnliTech) with monochromatic light focused on device pixels and a chopper frequency of 20 Hz. For tandem solar cells, the J - V characteristics were carried out under the illumination of a two-lamp high spectral match solar simulator (SAN-EI ELECTRIC, XHS-50S1). EQE measurements were performed in ambient air, and the bias illumination from highly bright LEDs with emission peaks of 850 and 460 nm was used for the measurements of the front and back subcells, respectively. No bias voltage was applied during the EQE measurements of tandems. The UPS measurements were performed using a Thermo SCIENTIFIC Nexsa system equipped with a He-I UV source (21.22 eV). The analysis spot size was $3 \text{ mm} \times 3 \text{ mm}$, and the base pressure in the analysis chamber was better than $2 \times 10^{-9} \text{ mbar}$. The instrument work function was calibrated to 4.2 eV. Depth profiling was conducted using an Ar⁺ ion gun with an etching energy of 2000 eV and medium current intensity. The sputtering rate was calibrated to 0.30 nm/s against a Ta₂O₅ standard. Data acquisition and processing were carried out using Thermo Advantage software (version 5.9921). The UPS system offers an energy resolution of $\leq 120 \text{ meV}$ and a sensitivity exceeding 2,200,000 cps. ToF-SIMS was performed using PHI nano TOF3. Samples were analyzed in the dual-beam proffling mode. All profiles were performed in the scatter mode. Spectral data were acquired in the high-mass resolution mode. The primary ion for analysis was 30 keV Bi³⁺⁺. This ion beam was applied over

a $50 \times 50 \mu\text{m}^2$ area at the center of the sputter crater. The sputter ion was 1 keV Ar^+ at 0.18 nm/s (with an Ar electron impact ion source). A $400 \times 400 \mu\text{m}^2$ sputter crater was used. KPFM measurements were performed using a Bruker Dimension Icon system. The KPFM mode was employed with SCM-PIT-V2 probes. The scanning frequency was 0.8 Hz, the lift height was set to 130 nm, and the image resolution was 256×256 pixels. KPFM measurements were conducted on a Bruker Dimension Icon system. The CPD values were subsequently derived from statistical analysis of the potential images using NanoScope Analysis software. GIWAXS measurements were performed using a Xeuss 2.0 instrument (France) equipped with a Pilatus3R 200K detector. The X-ray source was a Cu target ($\lambda = 0.154 \text{ nm}$). The scattering vector (q) range was from -2 to $2 \text{ \AA}^{-1}$. The sample-to-detector distance was set to 1,600 mm. GIWAXS patterns were acquired at incident angles of 0.05° and approximately 1° using a grazing-incidence sample stage.

Stability tests of solar cells

The operational stability tests were carried out under full AM1.5 G illumination (multi-colored LED solar simulator, 100 mW cm^{-2}) using a home-build LabVIEW-based MPP tracking system in ambient conditions with humidity of $\sim 40\%$. The modules were encapsulated with a cover glass and butyl rubber, POE padding is used in the middle. They were melted at 100°C and applied pressure to expel air. The dark long-term shelf stability assessments of modules were carried out by repeating the $J-V$ characterizations over various times, and the devices were stored in ambient conditions with humidity of $40 \pm 5\%$.

Other characterizations

SEM images were obtained using a TESCAN microscope with an accelerating voltage of 2 kV. XRD patterns were collected using a Bruker D8 Advance equipped with a NaI scintillation counter and using monochromatized Copper $K\alpha$ radiation ($\lambda = 1.5406 \text{ \AA}$). Photoluminescence (PL) was measured using a HORIBA Nanolog fluorescence spectrometer (Horiba Instruments) with an excitation wavelength of 485 nm. The intensity of the laser was adjusted to 1-Sun equivalent. TRPL was measured using a Horiba Fluorolog-3 time-correlated single-photon counting system; the samples were excited using a pulsed laser with a wavelength of 485 nm from the perovskite side. The PL decay curves were fitted with a single exponential function for perovskite films

fabricated on glass. The PL decay curves were fitted with biexponential function to obtain the fast and slow PL decay lifetimes of τ_1 and τ_2 and the corresponding coefficients of A_1 and A_2 of perovskite films, respectively. Then the PL effective decay lifetime τ_{eff} was calculated by the following equation: $\tau_{\text{eff}} = (A_1\tau_1 + A_2\tau_2)/(A_1 + A_2)$. The PLQY was obtained via exciting the corresponding samples with a 405 nm laser inside an integrated sphere (Newport, 70682NS). The signal was coupled into a fiber and collected by an Ocean Optics QEPro spectrometer. The integrating sphere and fiber were calibrated using a lamp of the known spectrum (Ocean Insight, HL-3P-INT-CAL). The illumination intensity was modulated with neutral density (ND) filters to achieve a near 1-sun equivalent photon flux at a bandgap of 1.8 eV. The PL imaging tool is an LIS-R1 system sourced from BT Imaging. The samples were excited with a continuous-wave 532 nm laser (one-sun on-sample intensity). The camera was a 1024 * 1024 pixels, charge-coupled-device silicon detector with a detection range of 400–1060 nm.

Data availability

Source data are provided with this paper. The main data supporting the findings of this study are available within the published article and its Supplementary Information and source data files. Additional data are available from the corresponding author on request.

References

1. Er-raji, O. et al. Electron accumulation across the perovskite layer enhances tandem solar cells with textured silicon. *Science* **390**, eadx1745 (2025).
2. Chin, Y. et al. Interface passivation for 31.25%-efficient perovskite/silicon tandem solar cells. *Science* **381**, 59-63 (2023).
3. Jia, L. et al. Efficient perovskite/silicon tandem with asymmetric self-assembly molecule. *Nature* **644**, 912-919 (2025).
4. Wu, W. et al. Stable and uniform self-assembled organic diradical molecules for perovskite photovoltaics. *Science* **389**, 195-199 (2025).
5. Wang, S. et al. Flexible perovskite/silicon tandem solar cells with 33.6% efficiency. *Nature* **649**, 59-64 (2025).

6. Fang, Z. et al. Flexible perovskite/silicon tandem solar cell with a dual buffer layer. *Nature* **649**, 65-72 (2025).
7. Chen, B. et al. Blade-coated perovskites on textured silicon for 26%-efficient monolithic perovskite/silicon tandem solar cells. *Joule* **4**, 850-864 (2020).
8. Liu, Z. et al. Strained heterojunction enables highperformance, fully textured perovskite/silicon tandem solar cells. *Joule* **8**, 2834-2850 (2024).
9. Fang, J. et al. Anion exchange promoting non-impurities enables conformable and efficient inverted perovskite solar cells. *Energy & Environmental Science* **17**, 7829-7837 (2024).
10. Wang, J. et al. Streamlined phase transition and reaction compensation in hybrid evaporation-solution deposited inverted perovskite solar cells. *Adv. Energy Mater.* **15**, 2404954 (2024).
11. Xiong, Z. et al. Homogenized chlorine distribution for >27% power conversion efficiency in perovskite solar cells. *Science* **390**, 638-642 (2025).
12. Liu, S. et al. Buried interface molecular hybrid for inverted perovskite solar cells. *Nature* **632**, 536-542 (2024).
13. Dunfield, S. et al. From defects to degradation: A mechanistic understanding of degradation in perovskite solar cell devices and modules. *Adv. Energy Mater.* **10**, 1904054 (2020).
14. Chen, R. et al. Reduction of bulk and surface defects in inverted methylammonium- and bromide-free formamidinium perovskite solar cells. *Nat. Mater.* **8**, 839-849 (2023).
15. Nie, R. et al. Enhanced coordination interaction with multi-site binding ligands for efficient and stable perovskite solar cells. *Nat. Commun.* **16**, 6438 (2025).
16. Zhou, J. et al. High-efficiency large-area perovskite solar cells via a multifunctional crystallization regulating passivation additive. *Adv. Mater.* **37**, 2502916 (2025).
17. Liu, Z. et al. Grain regrowth and bifacial passivation for high-efficiency wide-bandgap perovskite solar cells. *Adv. Mater.* **13**, 2203230 (2022).
18. Lu, H. et al. Vapor-assisted deposition of highly efficient, stable black-phase FAPbI₃ perovskite solar cells. *Science* **370**, 74 (2020).
19. Luo, X. et al. Efficient perovskite/silicon tandem solar cells on industrially compatible textured silicon. *Adv. Mater.* **35**, 2207883 (2023).

20. Tan, S. et al. Surface reconstruction of halide perovskites during post-treatment. *Journal of the American Chemical Society* **143**, 6781-6786 (2021).
21. Luo, D. et al. Enhanced photovoltage for inverted planar heterojunction perovskite solar cells. *Science* **360**, 1442-1446 (2018).
22. Wang, X. et al. Efficient and stable CsPbI₃ inorganic perovskite photovoltaics enabled by crystal secondary growth. *Adv. Mater.* **33**, 2103688 (2021).
23. Yang, M. et al. Facile fabrication of large-grain CH₃NH₃PbI_{3-x}Br_x films for high-efficiency solar cells via CH₃NH₃Br selective Ostwald ripening. *Nat. Commun.* **7**, 12305 (2016).
24. Hou, Y. et al. Efficient tandem solar cells with solution-processed perovskite on textured crystalline silicon. *Science* **367**, 1135-1140 (2020).
25. Xu, Q. et al. Diffusible capping layer enabled homogeneous crystallization and component distribution of hybrid sequential deposited perovskite. *Adv. Mater.* **36**, 2308692 (2023).
26. Du, S. et al. Bottom directional deposition perovskite heterojunctions for efficient and stable lead halide perovskite/silicon tandem solar cells. *Energy Environ. Sci.* **18**, 8827-8837 (2025).
27. Huang, T. et al. Performance-limiting formation dynamics in mixed-halide perovskites. *Sci. Adv.* **7**, eabj1799 (2021).
28. An, Y. et al. Optimizing crystallization in wide-bandgap mixed halide perovskites for high-efficiency solar cells. *Adv. Mater.* **36**, 2306568 (2023).
29. Chen, S. et al. Stabilizing perovskite-substrate interfaces for high-performance perovskite modules. *Science* **373**, 902-907 (2021).
30. Meggiolaro, D. et al. Formation of surface defects dominates ion migration in lead-halide perovskites. *Acs Energy Lett.* **4**, 779-785 (2019).
31. Tan, S. et al. Stability-limiting heterointerfaces of perovskite photovoltaics. *Nature* **605**, 268-273 (2022).
32. Yang, S. et al. Stabilizing halide perovskite surfaces for solar cell operation with wide-bandgap lead oxysalts. *Science* **365**, 473-478 (2019).
33. Wang, Q. et al. Scaling behavior of moisture-induced grain degradation in polycrystalline hybrid perovskite thin films. *Energy Environ. Sci.* **10**, 516-522 (2017).

34. Suo, J. et al. Multifunctional sulfonium-based treatment for perovskite solar cells with less than 1% efficiency loss over 4,500-h operational stability tests. *Nat. Energy* **9**, 172-183 (2024).
35. Li, Y. et al. 25.71 %-efficiency FACsPbI₃ perovskite solar cells enabled by a thiourea-based isomer. *Angew. Chem. Int. Ed.* **63**, e202410378 (2024).
36. Liu, Z. et al. All-perovskite tandem solar cells achieving >29% efficiency with improved (100) orientation in wide-bandgap perovskites. *Nat. Mater.* **24**, 252-259 (2025).
37. Wang, Y. et al. Improved solvent systems for commercially viable perovskite photovoltaic modules. *Science* **390**, 1021-1028 (2025).
38. Li, M. et al. In situ coating strategy for flexible all-perovskite tandem modules. *Nat. Photonics* **19**, 1255-1263 (2025).
39. Wen, J. et al. Buried heterointerface reinforcement with passivation-integrated nanostructures for efficient and stable perovskite solar modules. *Joule* **9**, 102212 (2025).
40. Lin, R. et al. All-perovskite tandem solar cells with dipolar passivation. *Nature* **648**, 600-606 (2025).
41. Farag, A. et al. Mitigation of open-circuit voltage losses in perovskite solar cells processed over micrometer-sized-textured Si substrates. *Adv. Funct. Mater.* **33**, 2210758 (2022).
42. Xie, G. et al. Crystallization thermodynamics regulation of 1.85 eV wide-Bandgap perovskite for efficient and stable perovskite-organic tandem photovoltaics. *Angew. Chem. Int. Ed.* **64**, e202501764 (2025).
43. Li, X. et al. Surface reconstruction for efficient and stable monolithic perovskite/silicon tandem solar cells with greatly suppressed residual strain. *Adv. Mater.* **35**, 2211962 (2023).
44. Jiang, X. et al. Isomeric diammonium passivation for perovskite–organic tandem solar cells. *Nature* **635**, 860-866 (2024).
45. Tan, Z. et al. Enhancing interfacial contact for efficient and stable inverted perovskite solar cells and modules. *Adv. Funct. Mater.* **35**, 2419133 (2025).

46. Fang, Z. et al. Surface reconstruction of wide-bandgap perovskites enables efficient perovskite/silicon tandem solar cells. *Nat. Commun.* **15**, 10554 (2024).

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Author contributions

Conceptualization, W.K., and H.T.; investigation, W.K., H.L., B.Y., and J.S.; resources, X.Q., R.Li., T.W., J.H., X.H., X.Z., H.F., L.Z., Z.C., M.L., R.L., W.Z., and H.T.; validation, W.K.; visualization, W.K.; writing - original draft, W.K.; writing - review & editing, B.Y., J.S., and H.T.; supervision, W.K. and H.T.; project administration, W.K. and H.T.; funding acquisition, W.K. and H.T.

Competing interests

Hairen Tan is the founder, Chief Scientific Officer, and Chairman of Renshine Solar Co., Ltd., a company that is commercializing perovskite PVs. The remaining authors declare no competing interests.

Fig. 1 | Schematic illustration of the hybrid sequential deposition process and cross-sectional SEM images of inorganic framework and perovskite film. **a** The deposition of organic salts with the mixed solution of FAI, FABr, MASCN and MACl. Followed by a continuous growth via annealing process. Scale bar, 2 μm . **b** The healing intervention of the as-prepared perovskite film with MASCN solution. Scale bar, 2 μm . All the above schematic diagrams were created using Microsoft Office PowerPoint 2021.

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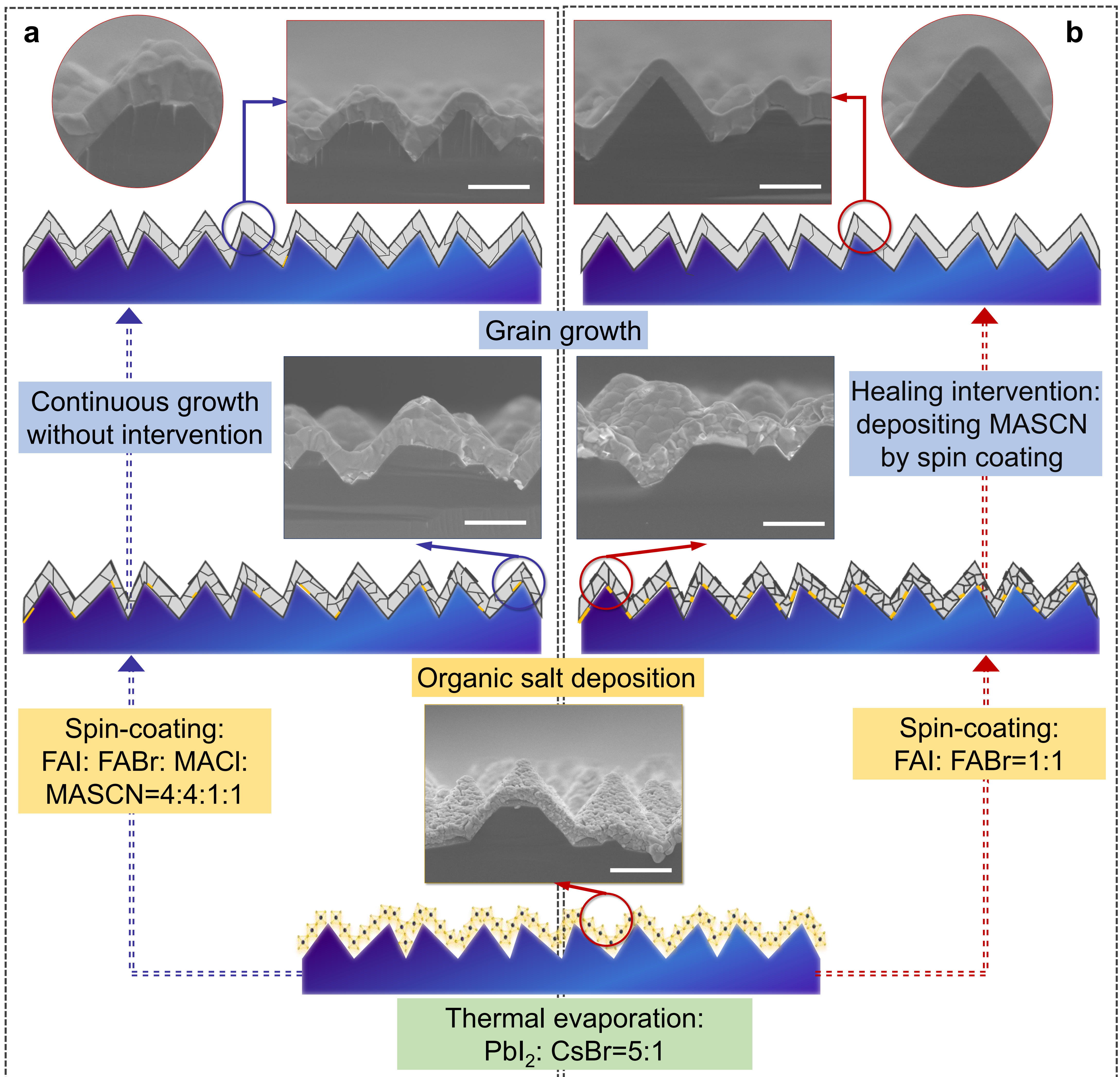
Fig. 2 | In situ characterizations of the MASCN healing intervention perovskite films. Top-view SEM images of **a** control and **b** target perovskite film on silicon during organic salt deposition and pre-annealing stage. The scale bars are 2 μm in **a**, and the scale bars are 3 μm in **b**. XRD patterns of the **c** control and **d** target film during organic salt deposition and pre-annealing stage. In situ PL spectra of **e** control and **f** target perovskite films during organic salt deposition and pre-annealing stage under a 405-nm laser illumination in a glove box. The colour scale shows intensity from low (blue) to high (red). **g** PL wavelength of control and target perovskite films extracted from **e** and **f** with times during organic salt deposition and pre-annealing stage.

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Fig. 3 | Characterization of perovskite films. **a** ToF-SIMS of control and target film deposited on silicon substrates. **b** Two-dimensional (2D) GIWAXS patterns of the control and target perovskite films on silicon (the incident angle of the X-ray beam is 1°). The colour scale shows

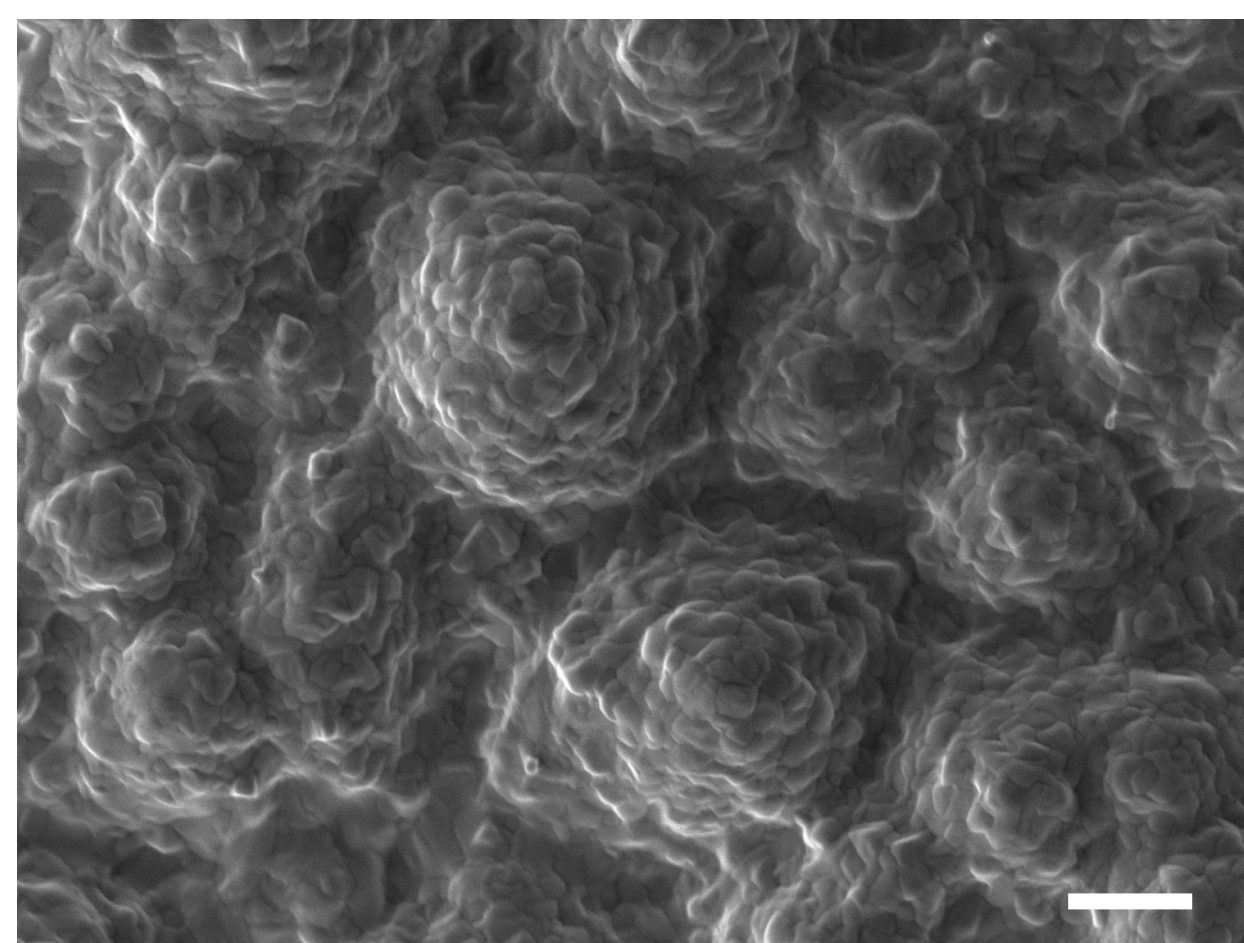
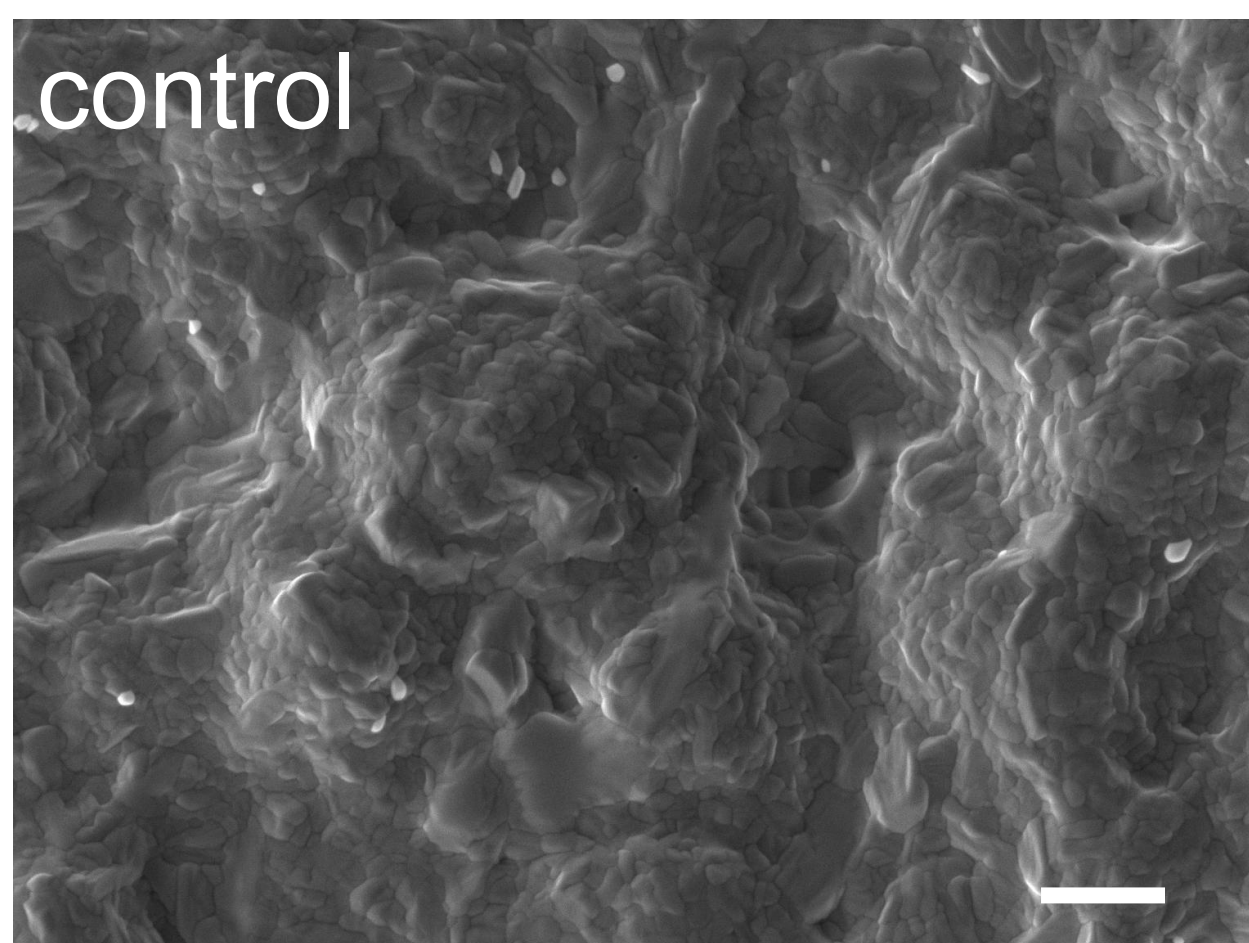
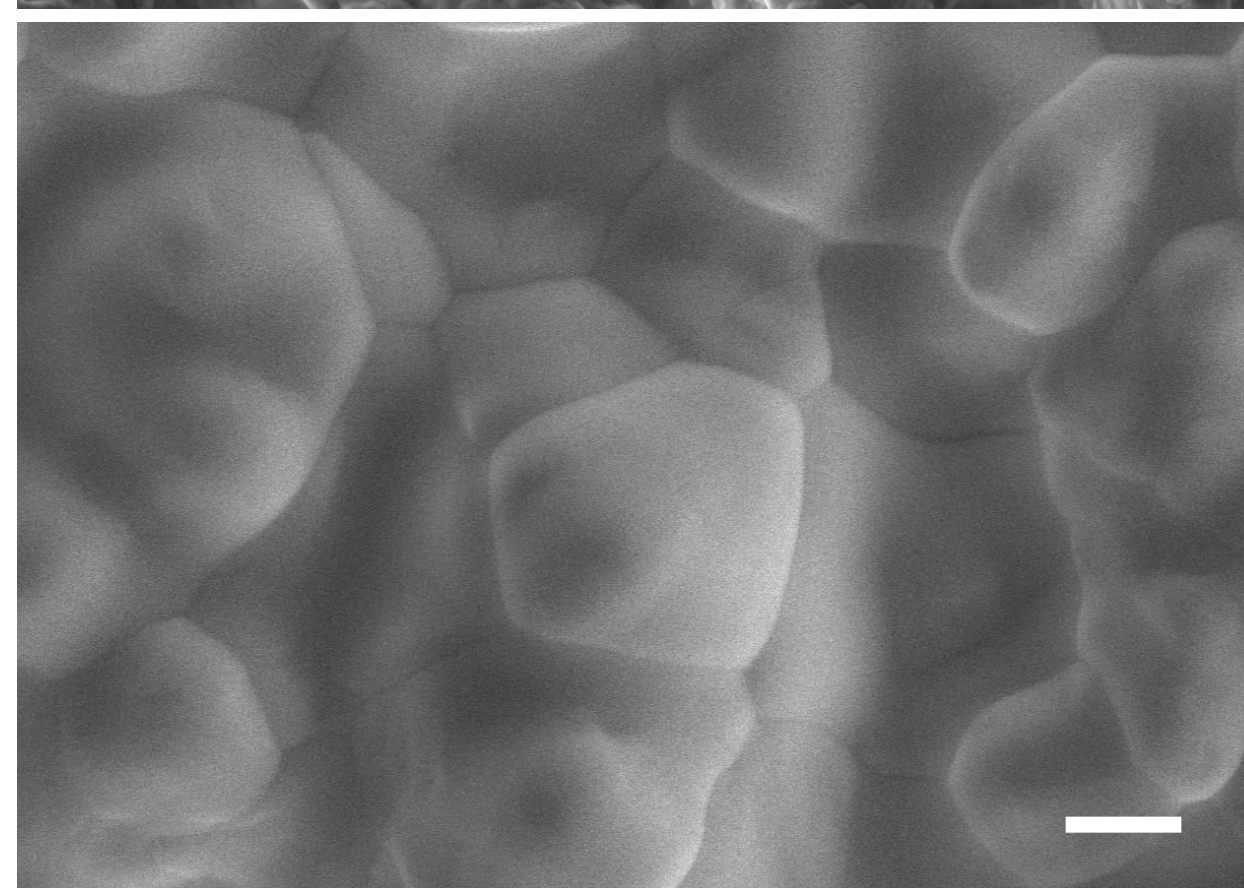
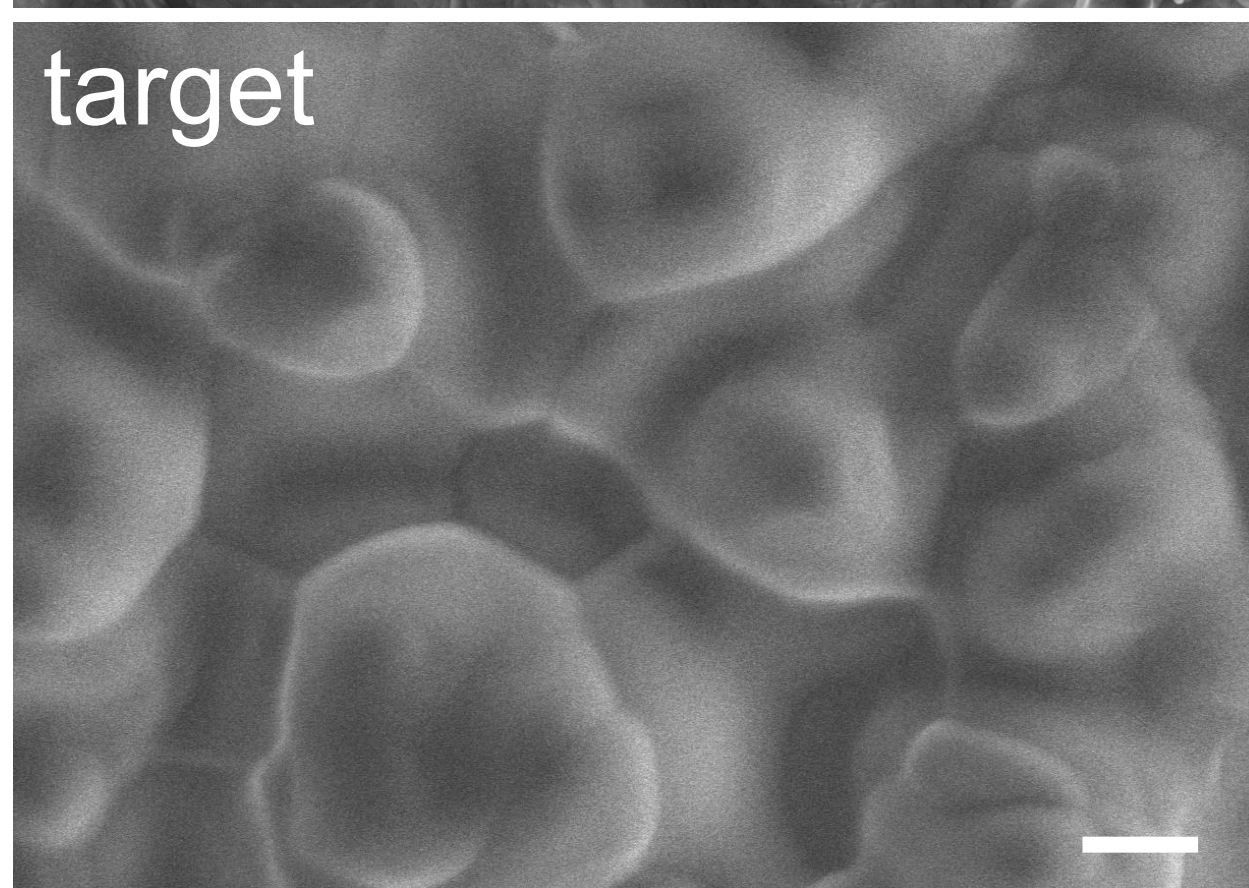
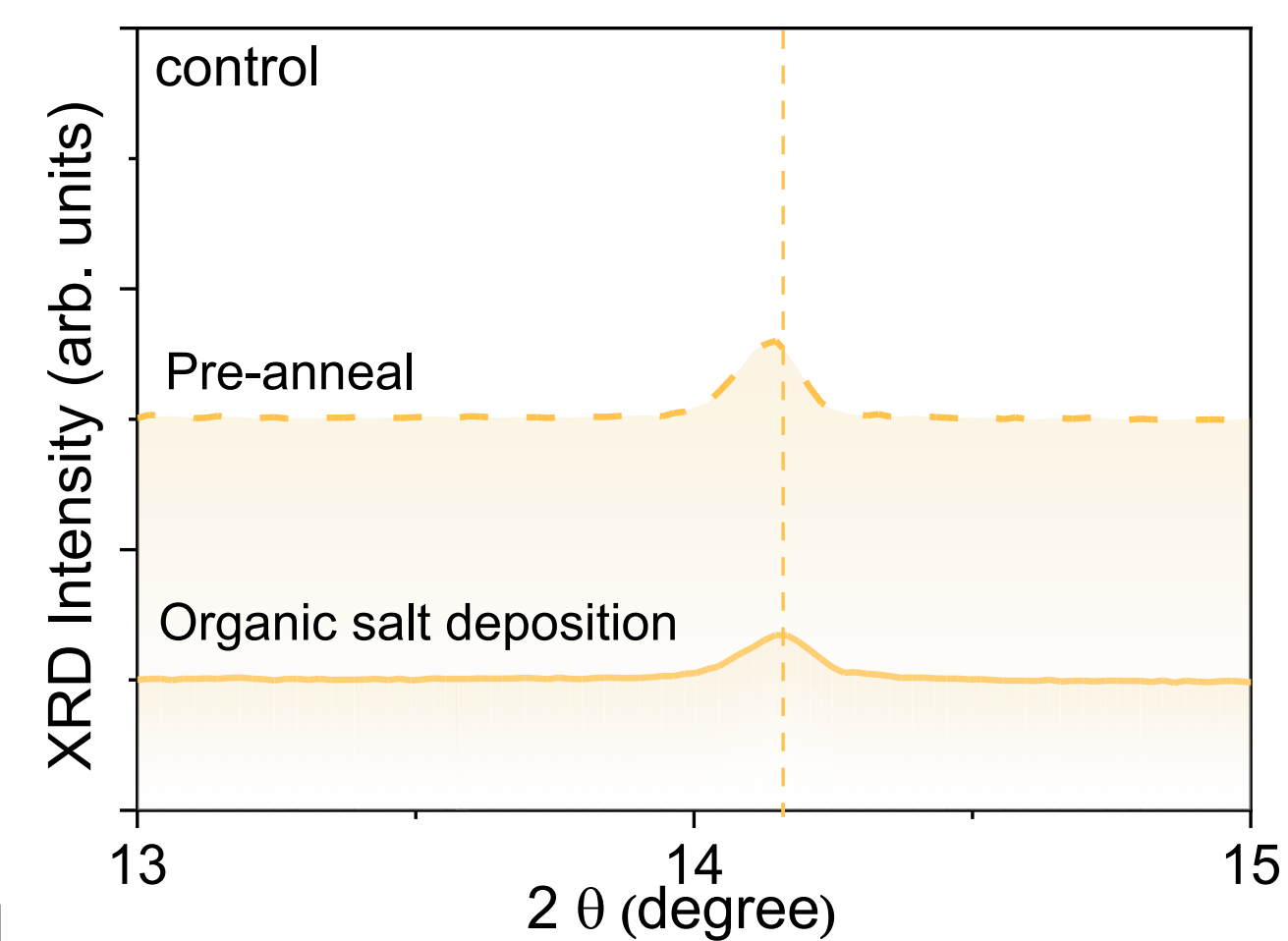
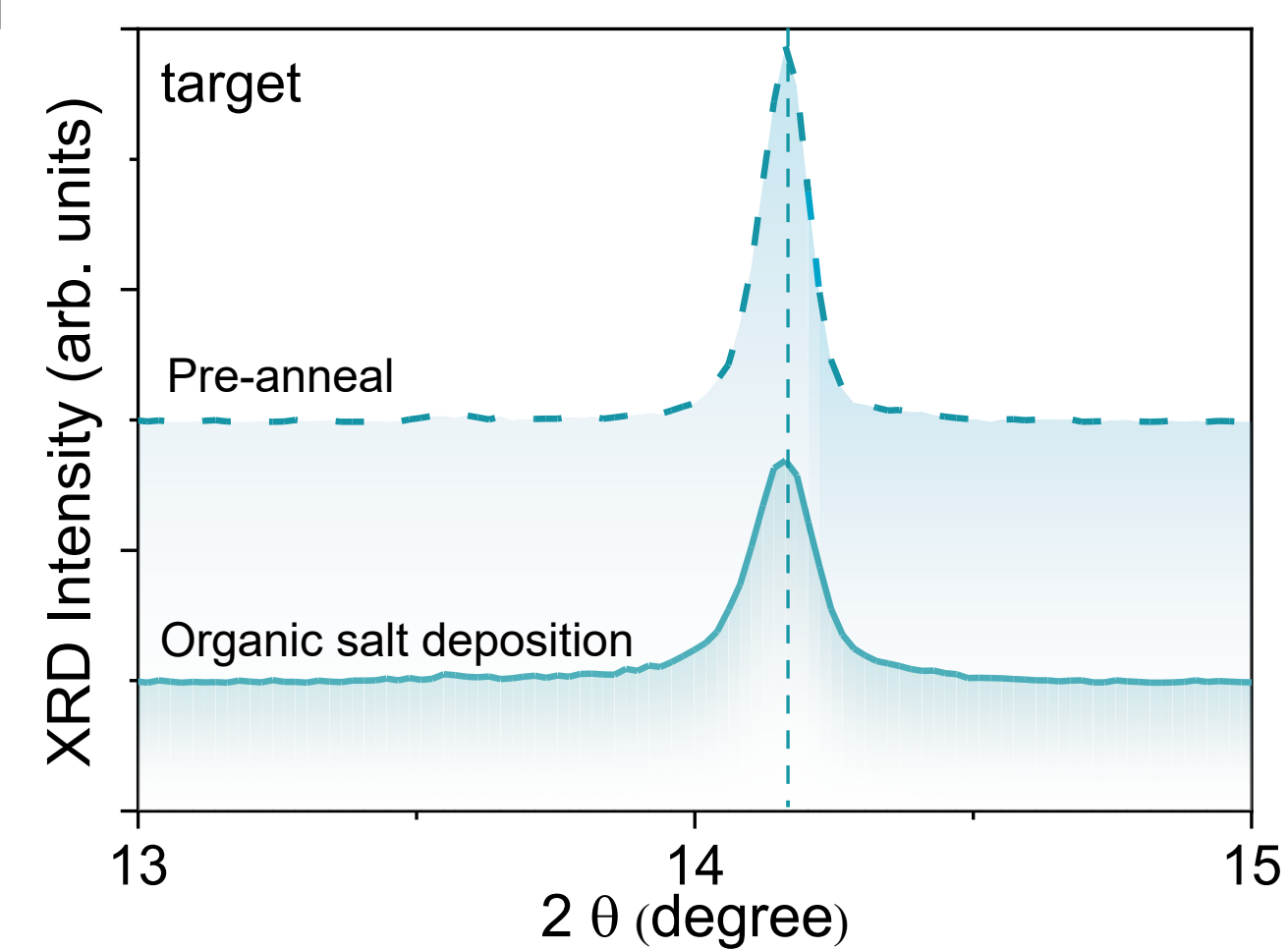
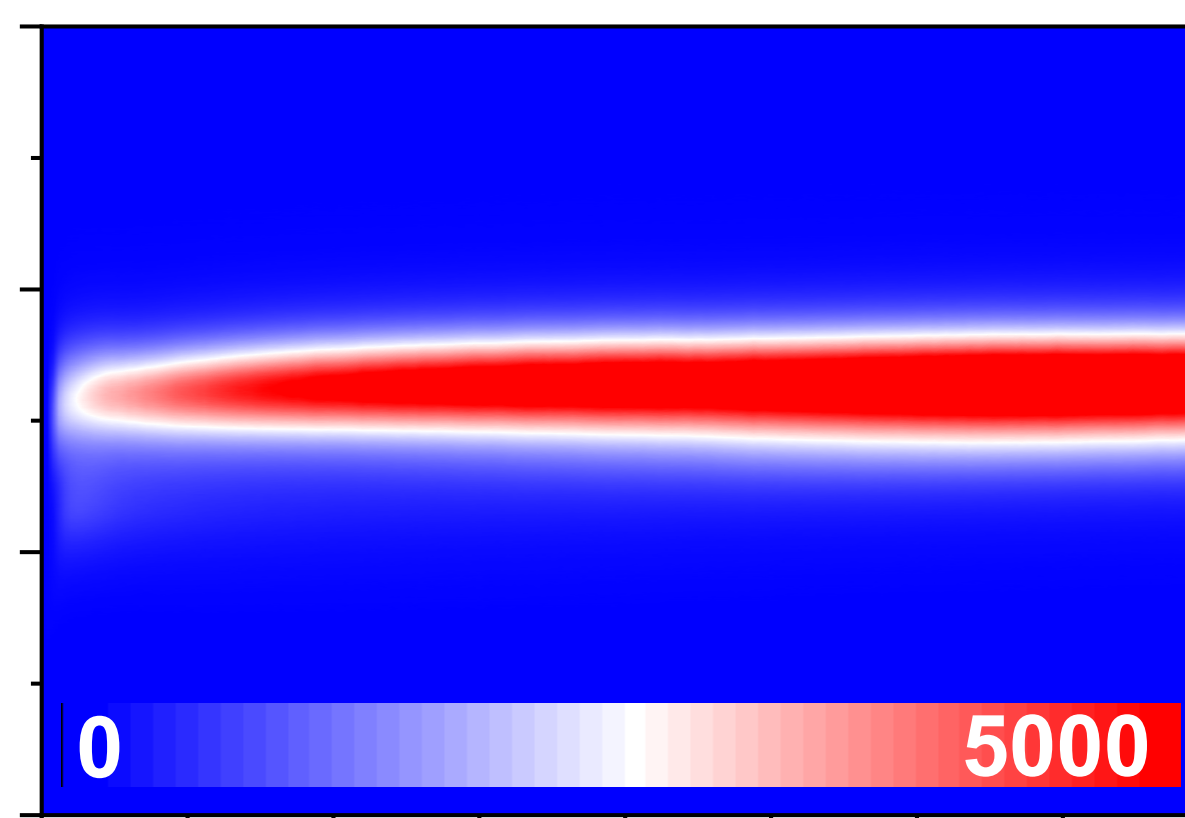
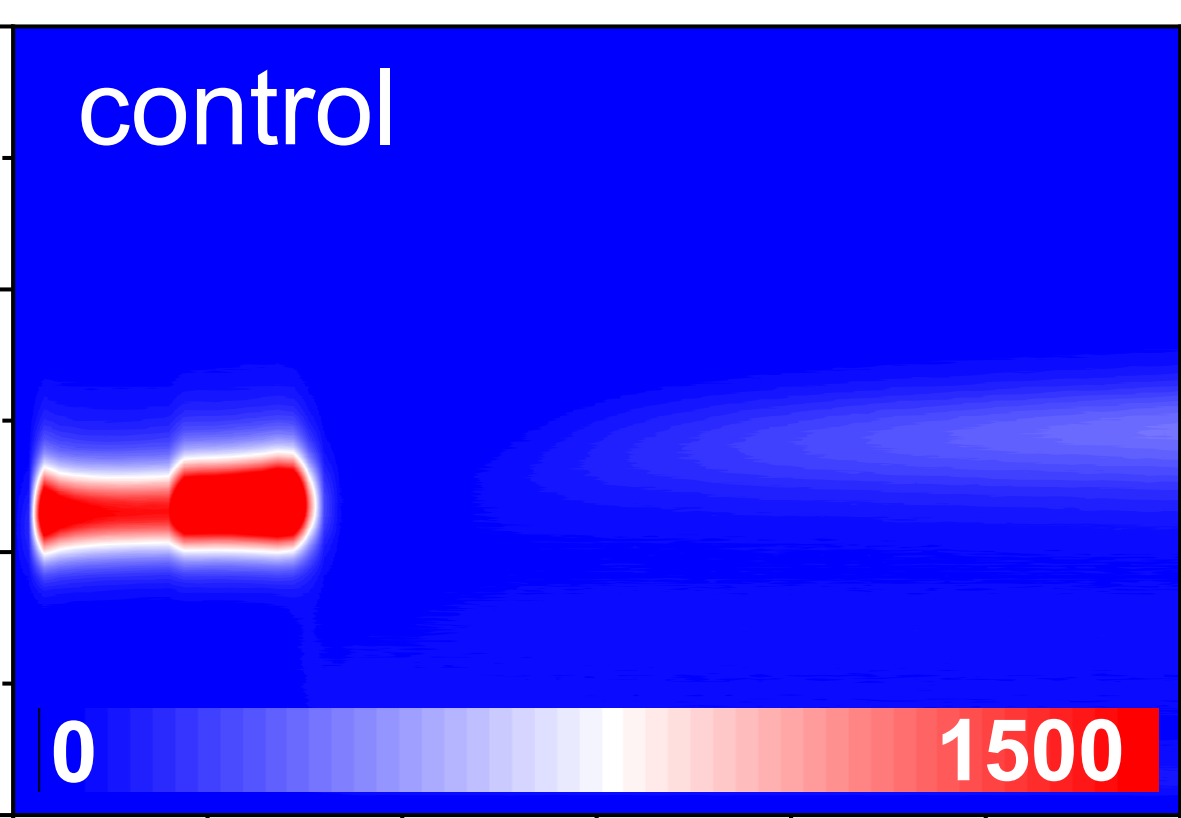
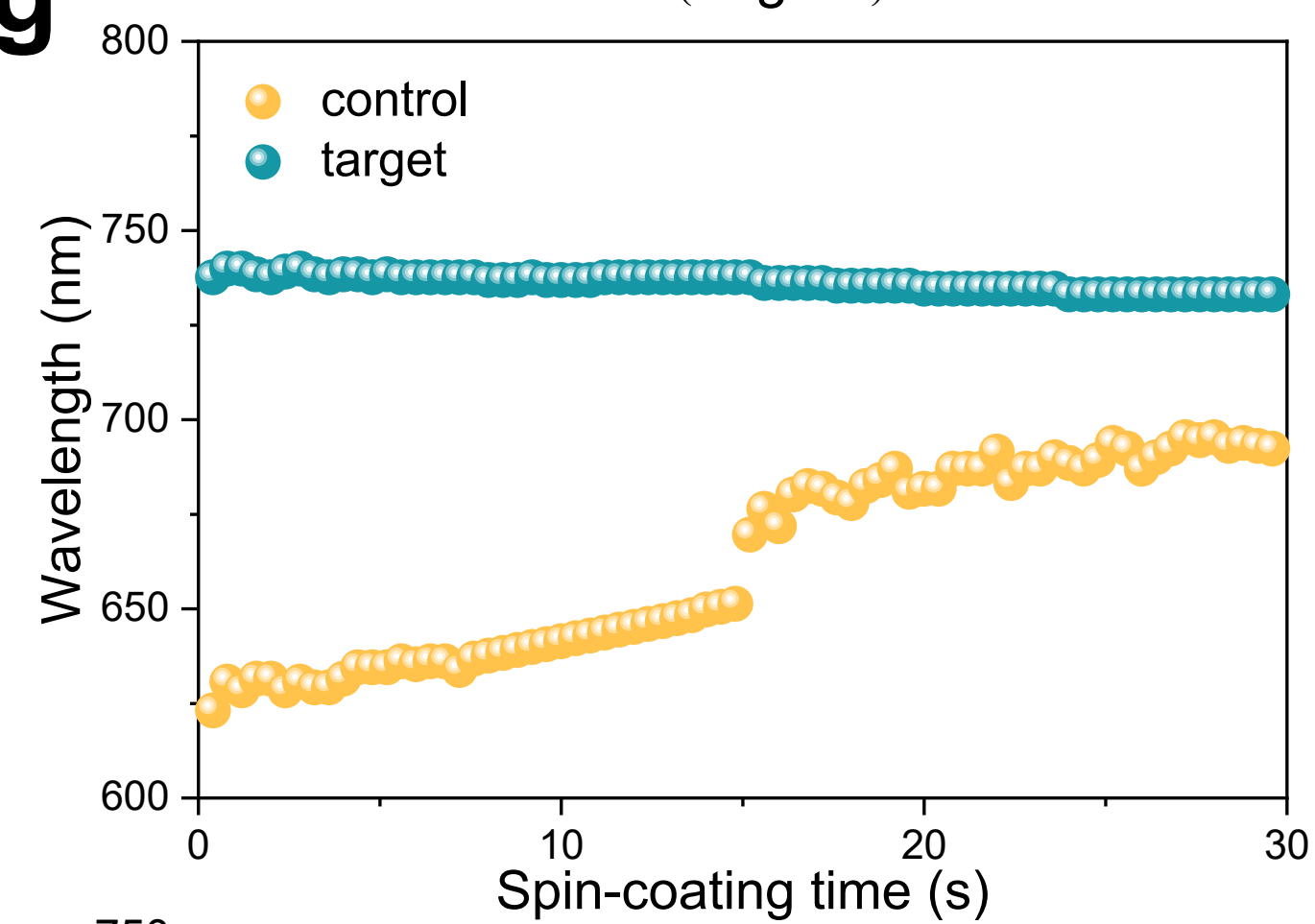
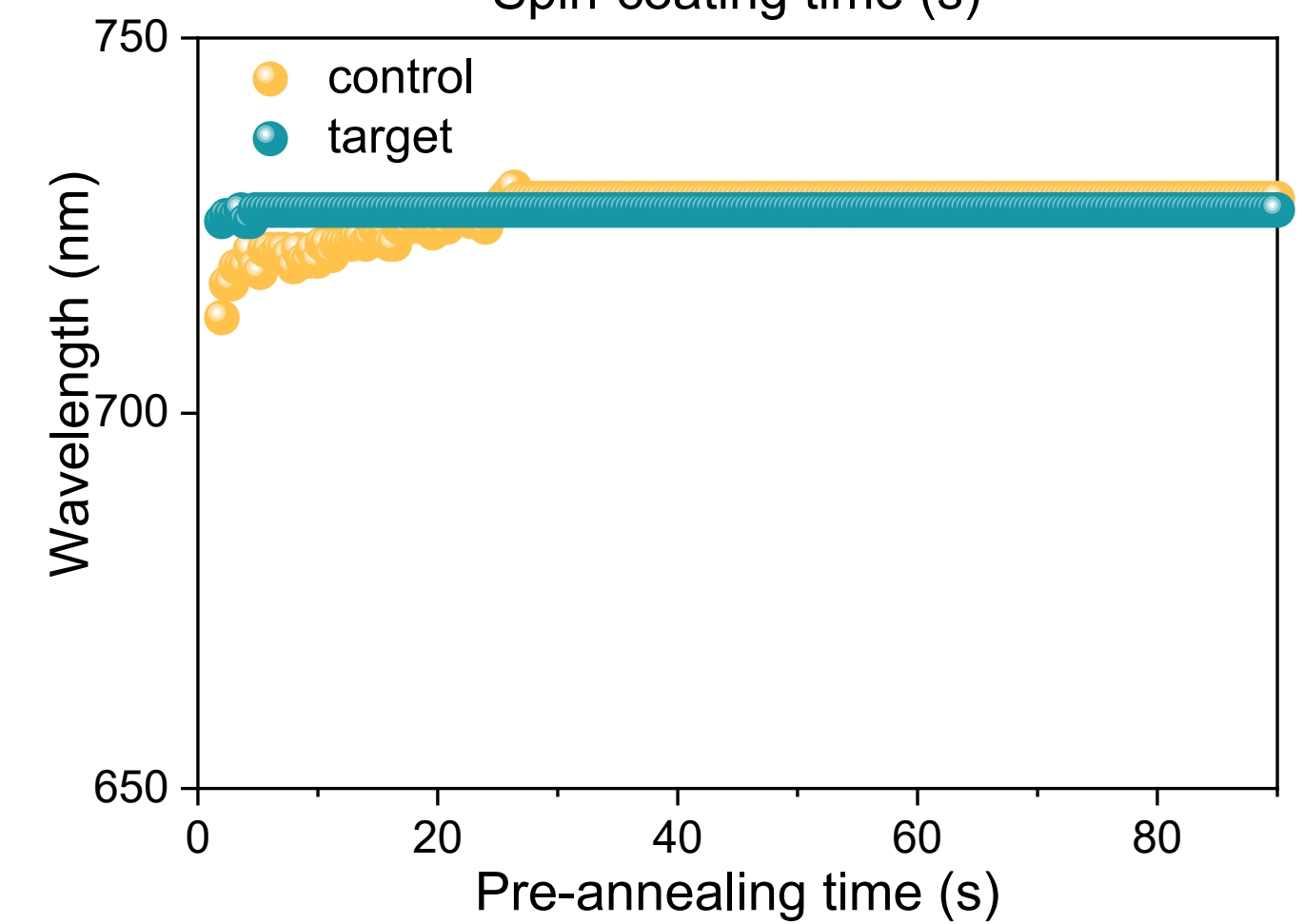
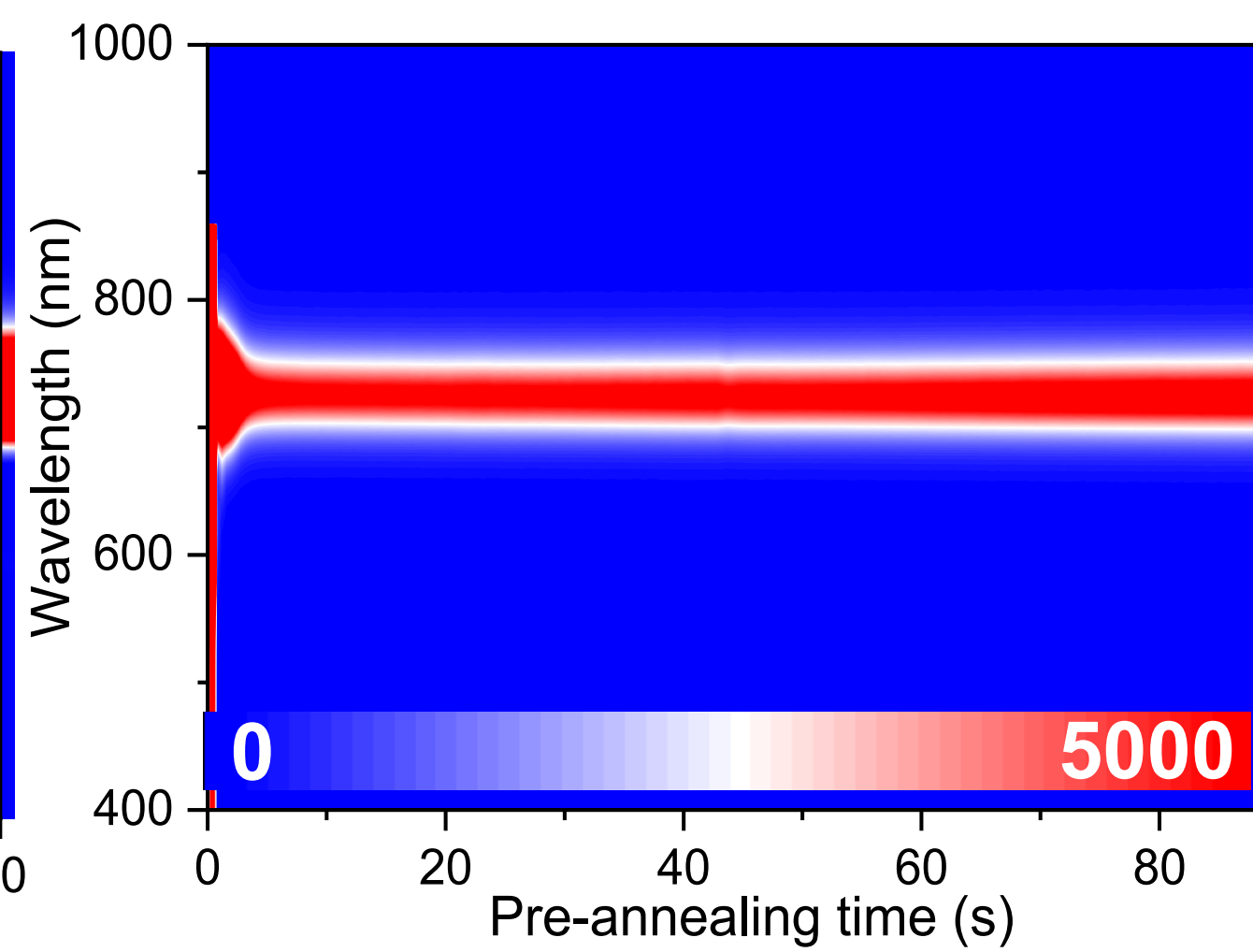
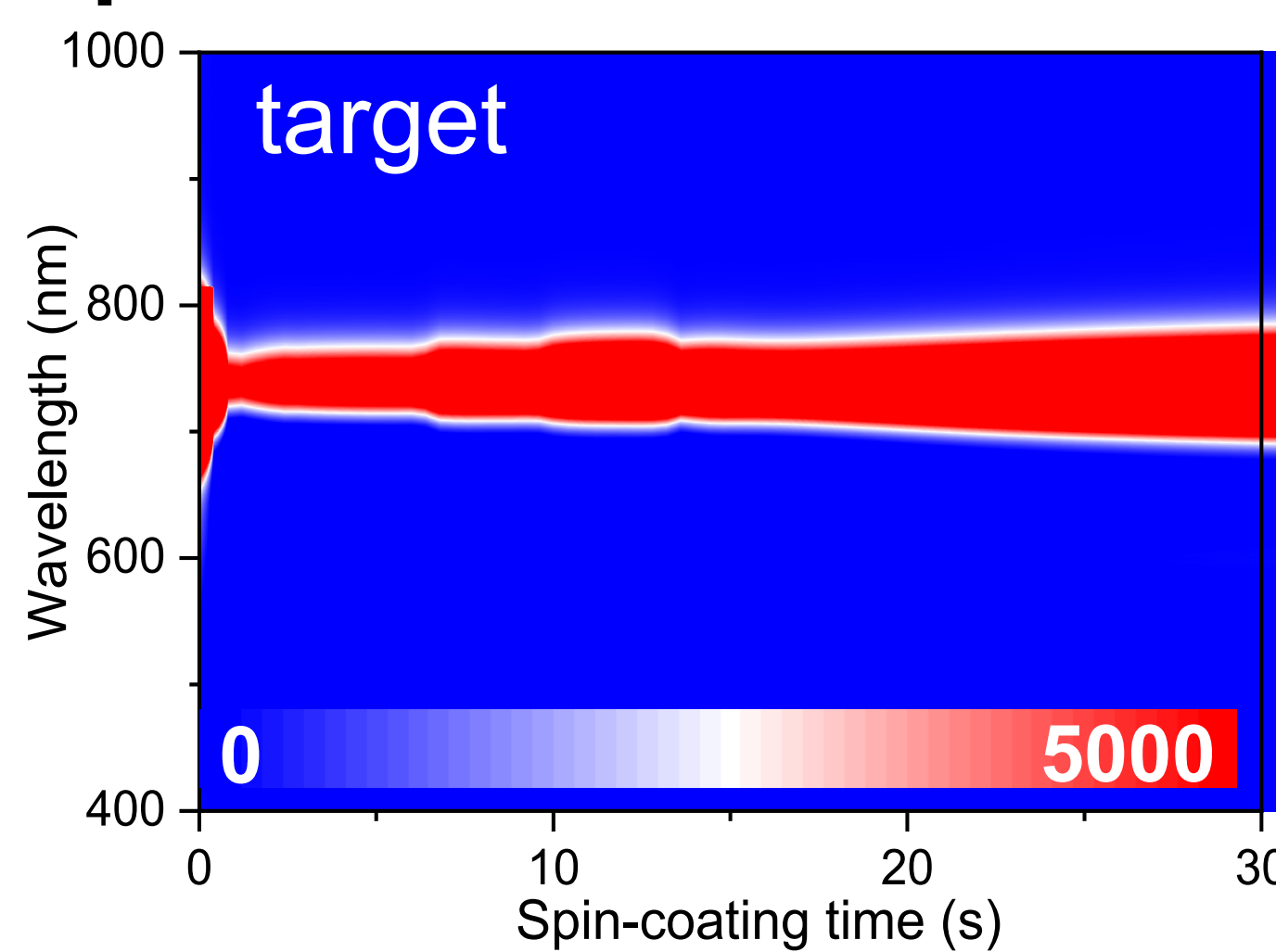
intensity from low (blue) to high (red). **c** PL mapping of the control and target perovskite films on silicon with the active area for $0.75\text{ cm} \times 1.5\text{ cm}$. The colour scale shows intensity from low (red) to high (yellow). **d** Steady-state PL spectra and **e** TRPL spectra of the control and target perovskite films on silicon. **f** QFLS values of control and target perovskite films on silicon. The standard deviation has been employed to assess data dispersion in statistical analyses. **g** Comparison of photovoltaic performance device between the control and target single-junction perovskite solar cells in the same runs. The box lines indicate the standard deviation, and the centre represents the mean value. The standard deviation has been employed to assess data dispersion in statistical analyses. **h** The best forward and reverse J - V scans of the target single-junction perovskite solar cells.

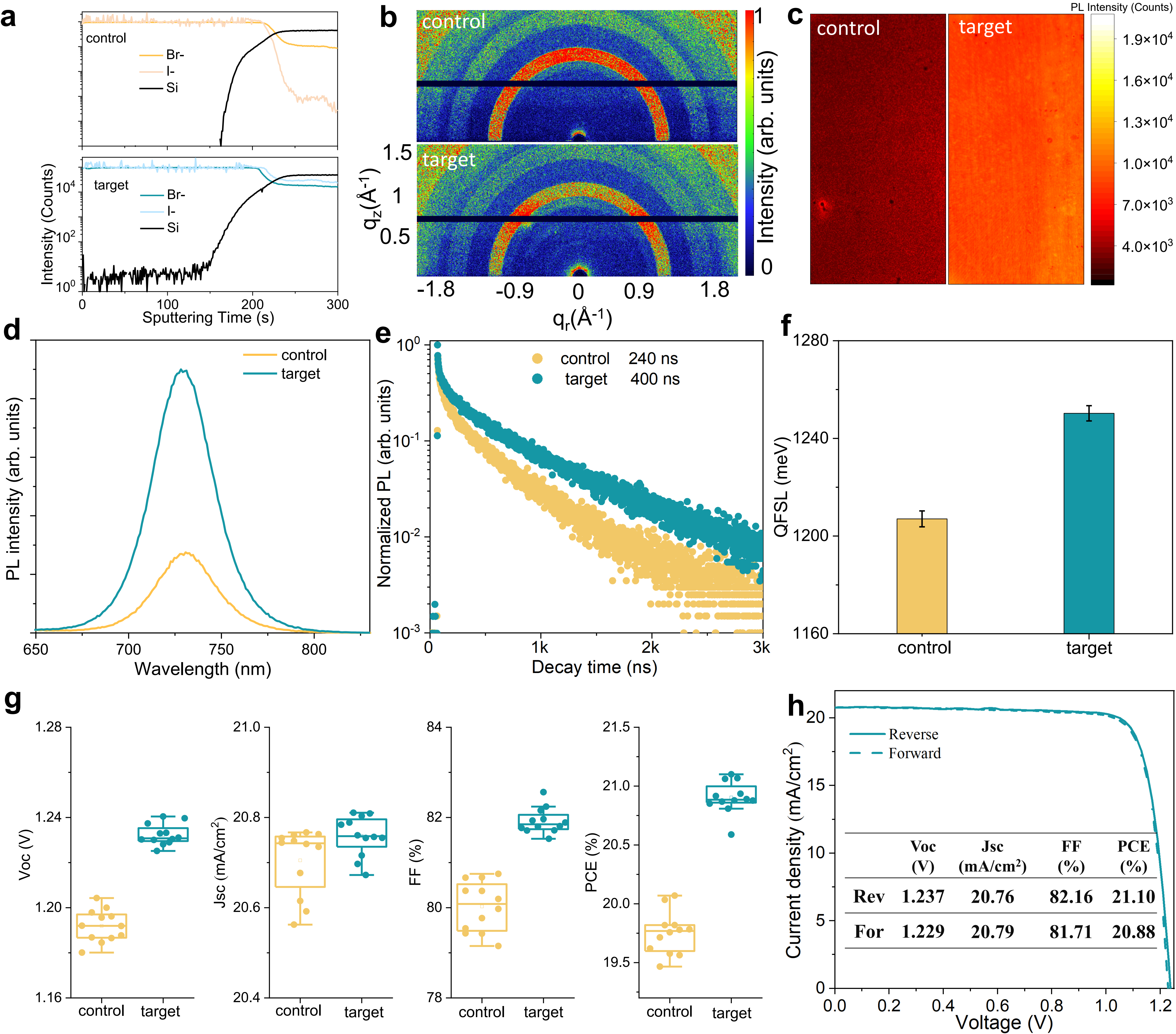
Fig. 4 | Device performance and stability. **a** Schematic illustration of the perovskite/silicon tandem architecture. The schematic diagram was created using Microsoft Office PowerPoint 2021. **b** The cross-section SEM images of the target tandem cells on textured silicon. Scale bar, 2 μm . **c** Comparison of photovoltaic performance between the control and target tandem cells in the same runs. The box lines indicate the standard deviation, and the centre represents the mean value. The standard deviation has been employed to assess data dispersion in statistical analyses. **d** The certified forward and reverse J - V scans of unencapsulated the target tandems. **e** EQE spectra and integrated J_{sc} for the target tandem devices. **f** The certified steady-state power output at the MPP under an illumination power of 100 mW/cm^2 for the target tandem devices. The inset in **f** is a photo of the target tandem device. **g** Continuous MPP stability testing for an encapsulated target tandem device under simulated 1-sun illumination at room temperature in air.

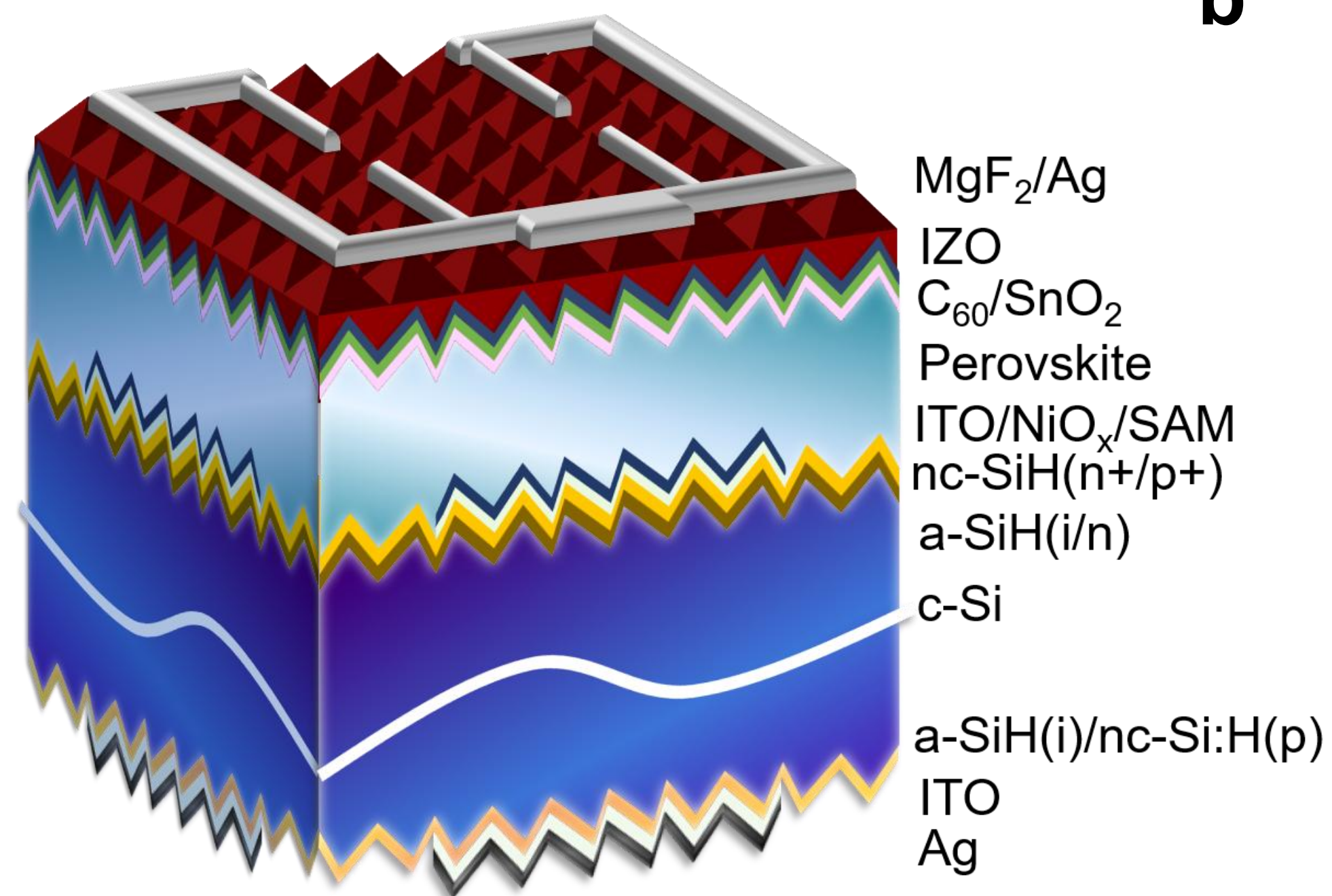
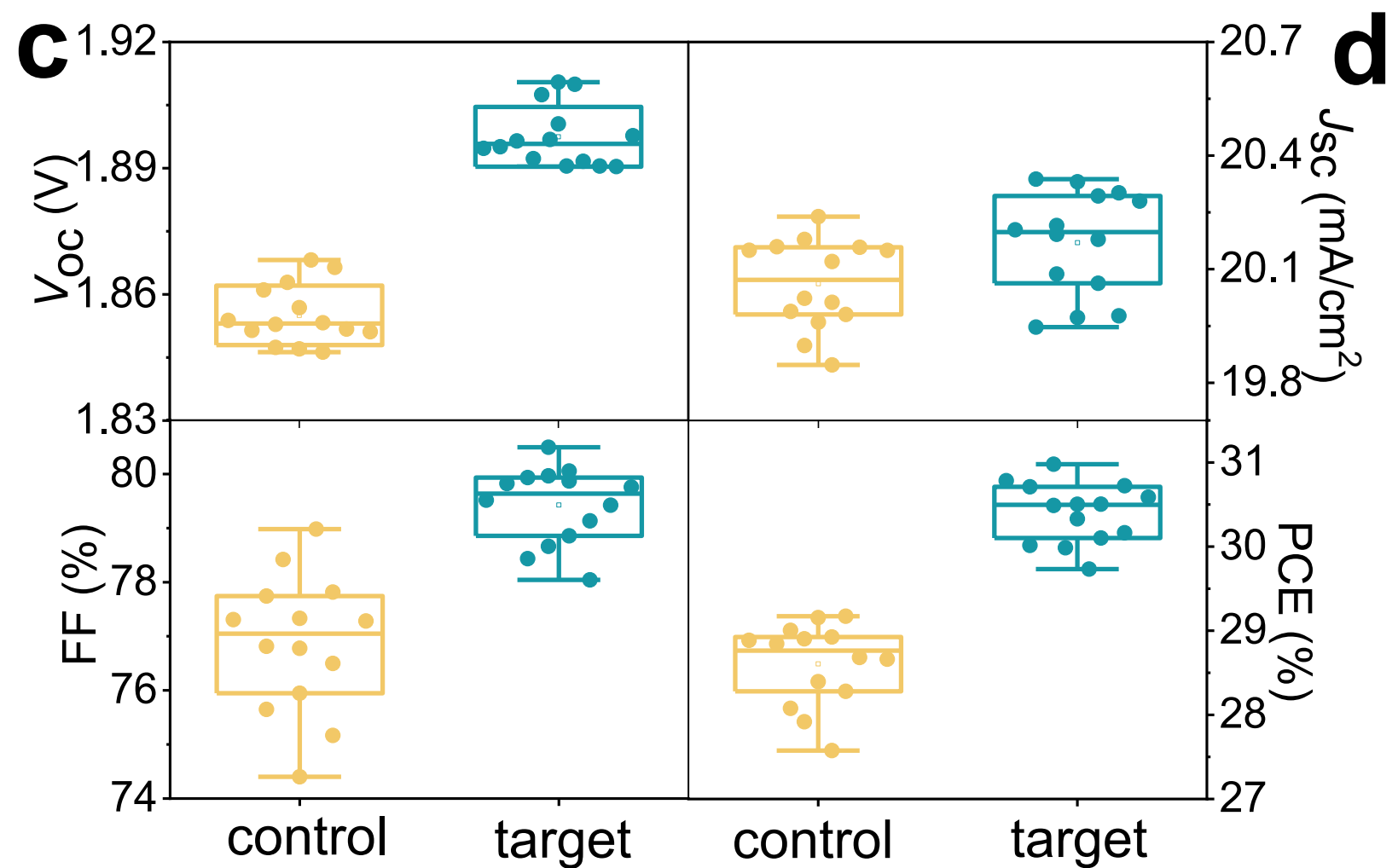
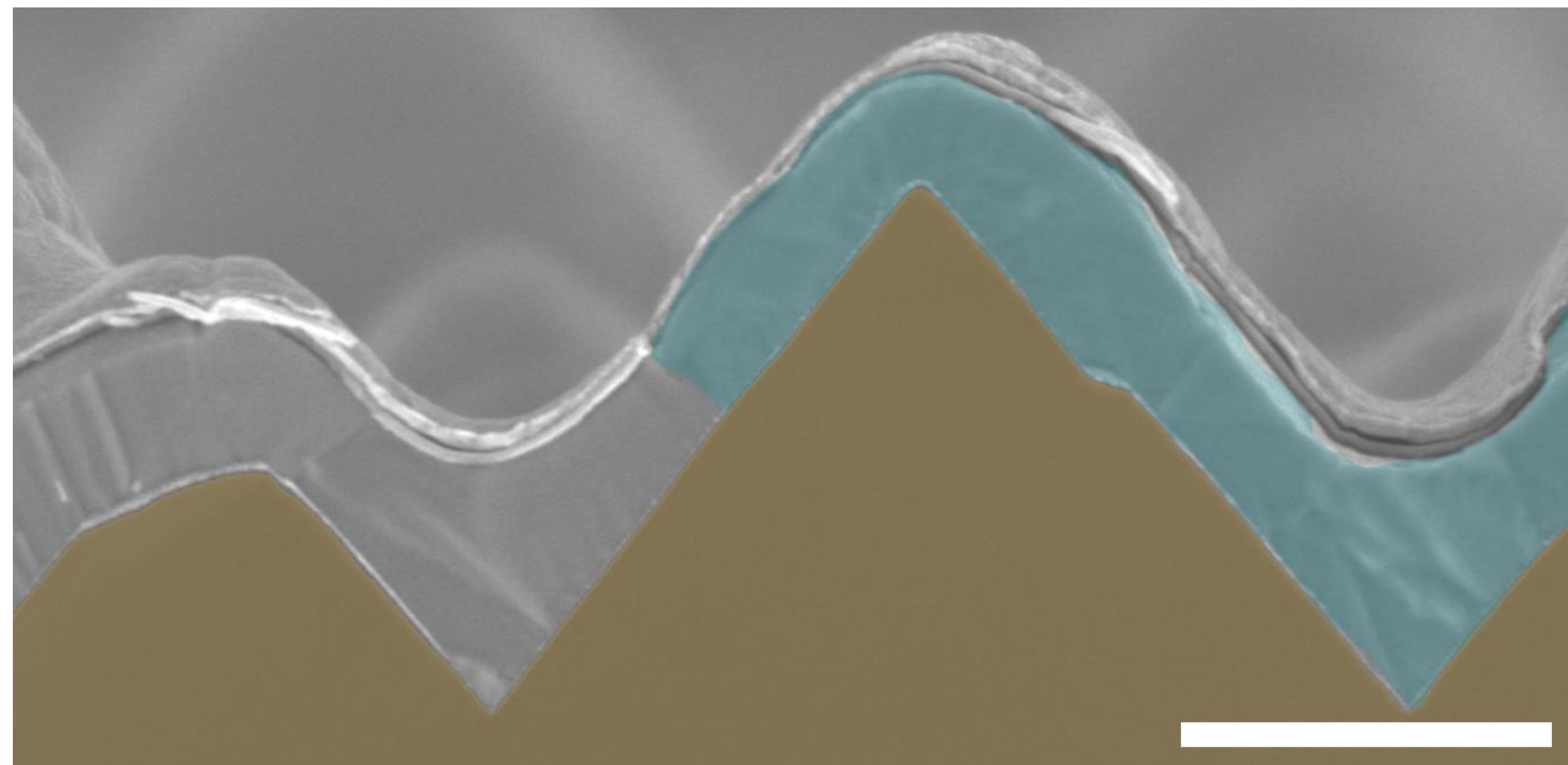
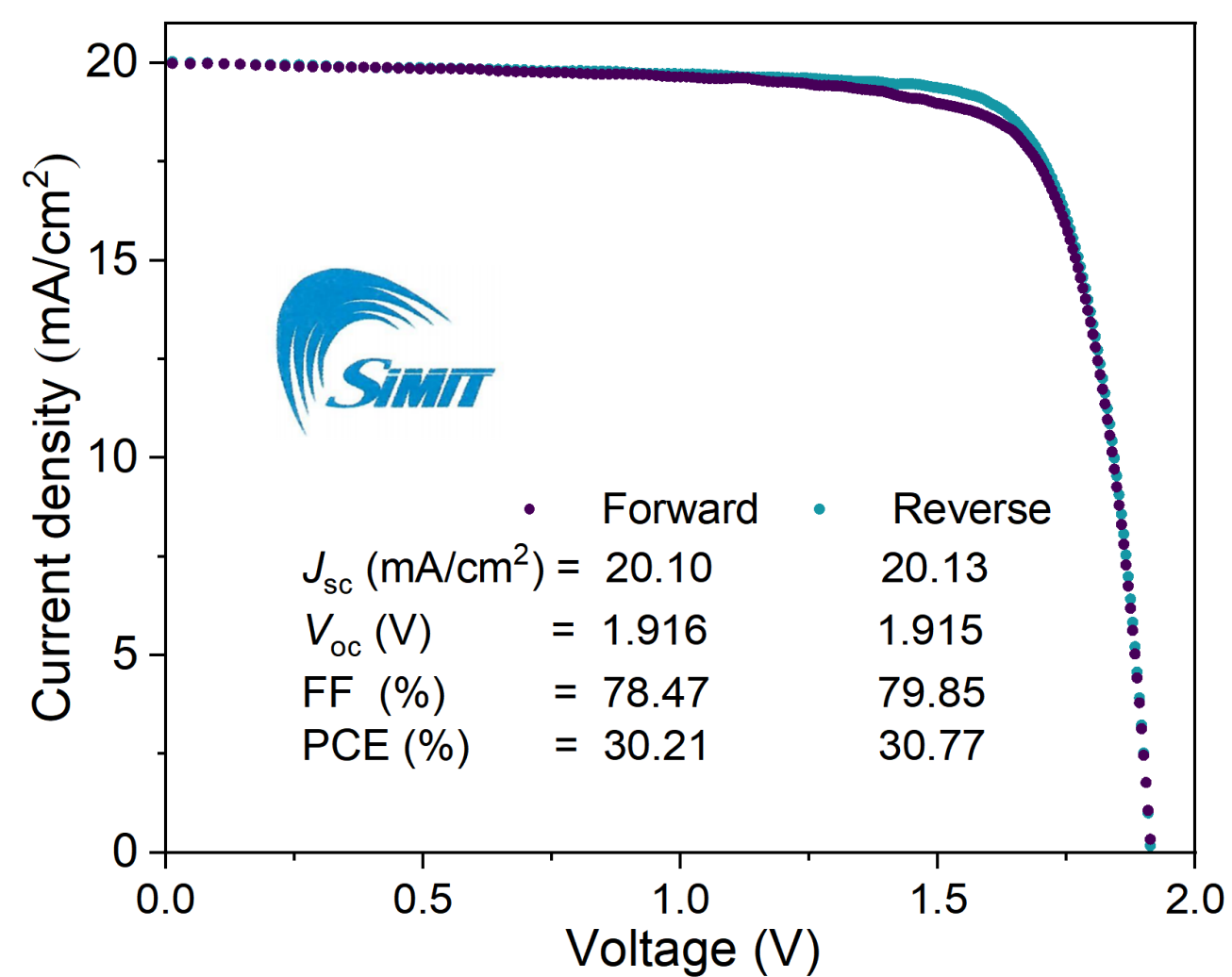
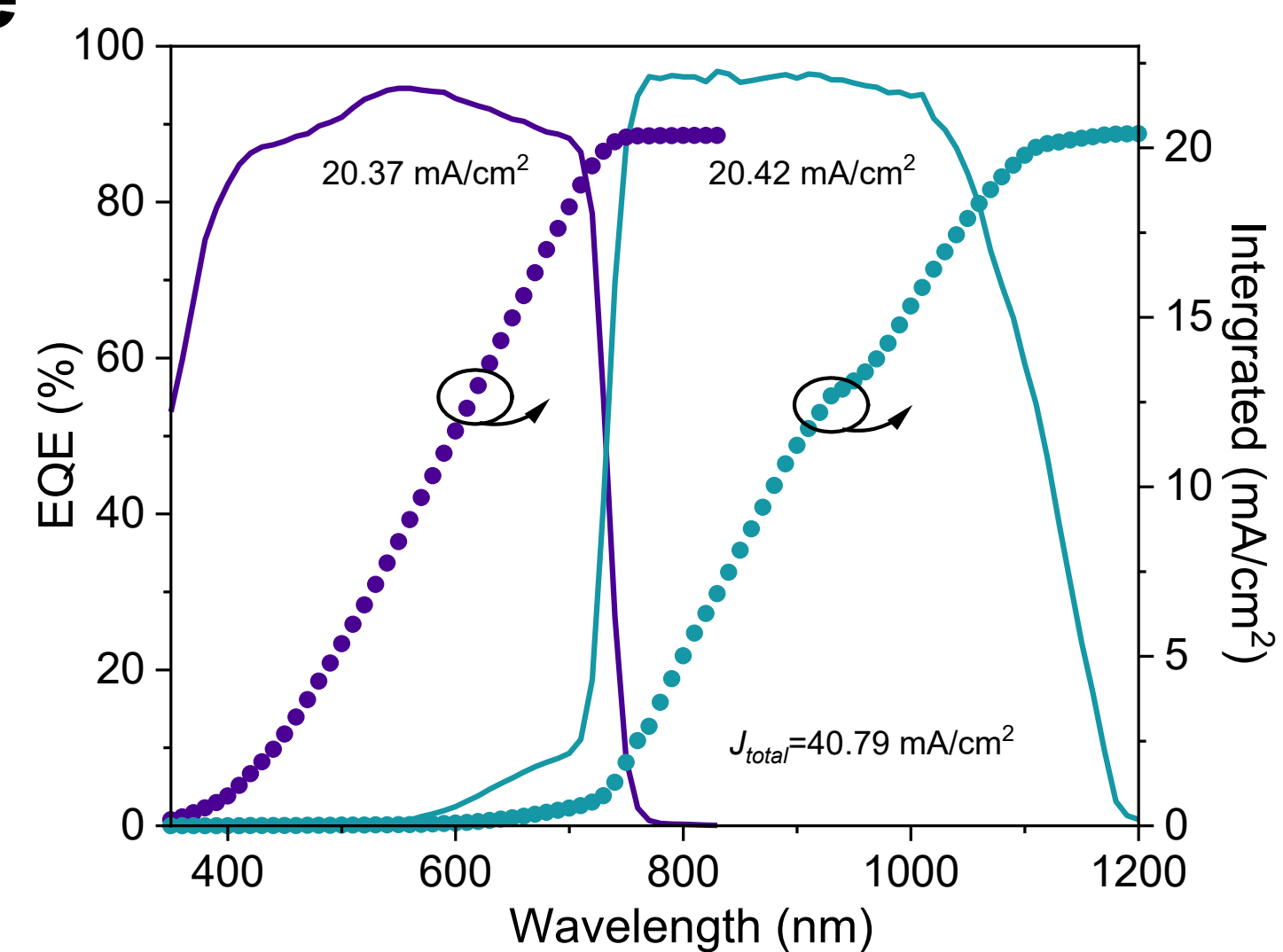
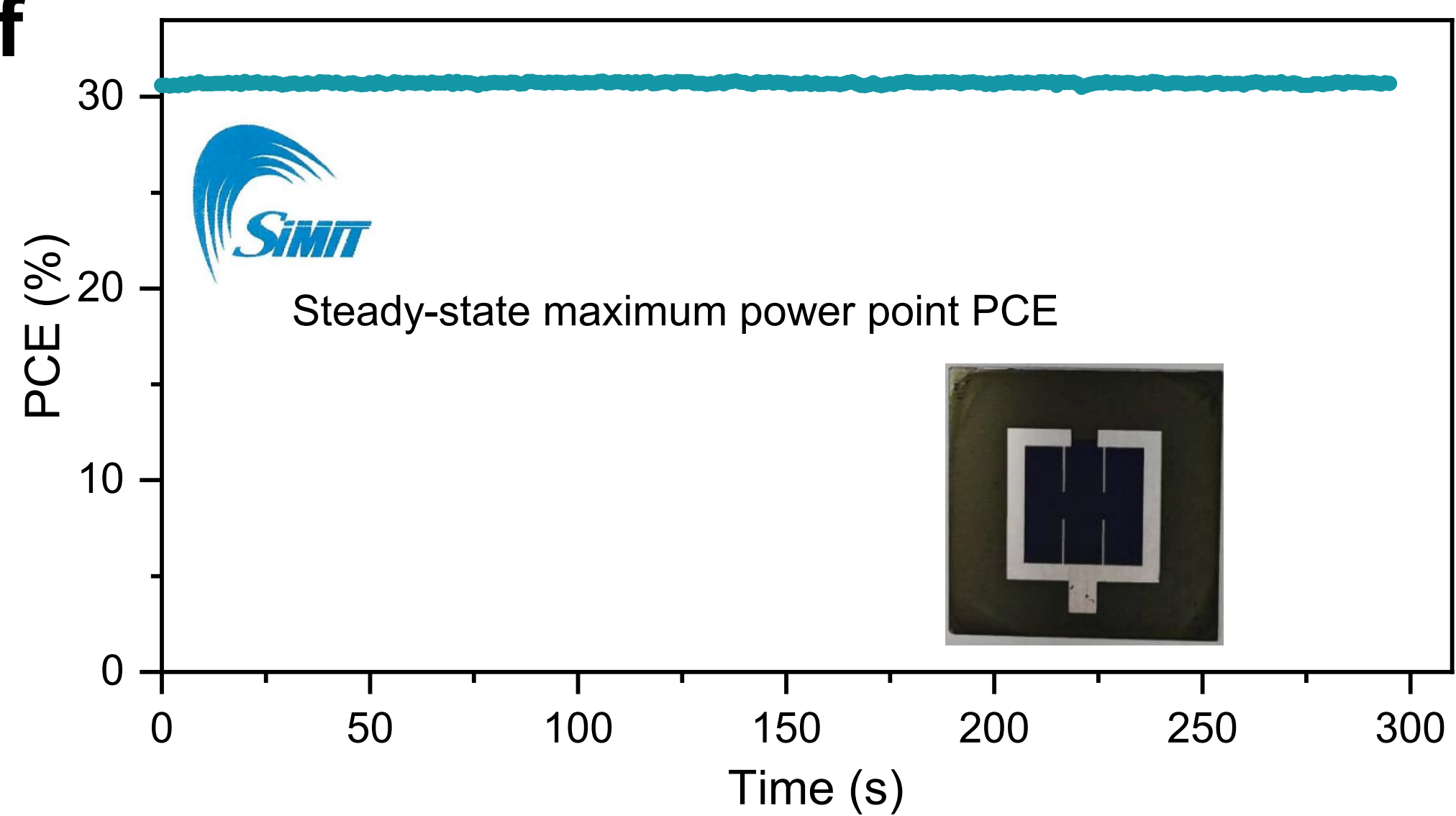


a Organic salt deposition

Pre-anneal

**b** target**c****d****e** control**g****f** target



a**b****d****e****f****g**